

4 Techniques of Integration

Definition 4.1 (Integration). *Integration is the reverse process of differentiation.*

Suppose $\frac{dy}{dx} = f(x)$. To obtain y , we integrate the function $f(x)$ with respect to the independent variable x . This is put in notation form as

$$y = \int f(x)dx + C,$$

where C is a constant of integration and $f(x)$ is the *integrand*. The following are the techniques of integration.

4.1 Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad \text{for } n \neq -1,$$

where C is a constant of integration.

Example(s):

$$(a) \int x^3 dx = \frac{x^{3+1}}{3+1} + C = \frac{x^4}{4} + C.$$

$$(b) \int x^{-7} dx = \frac{x^{-7+1}}{-7+1} + C = -\frac{x^{-6}}{6} + C.$$

$$(c) \int \sqrt{x} dx = \int x^{1/2} dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C = \frac{2}{3}x^{3/2} + C.$$

$$(d) \int x(1-3x)dx = \int (x-3x^2)dx = \frac{x^2}{2} - 3 \cdot \frac{x^3}{3} = \frac{x^2}{2} - x^3 + C.$$

$$(e) \int (2x-3)^2 dx = \int (4x^2 - 12x + 9)dx = \frac{4}{3}x^3 - 6x^2 + 9x + C.$$

(f)

$$\begin{aligned} \int \frac{\sqrt{x} + \sqrt[3]{x} + 6}{x^2} dx &= \int \left(\frac{x^{1/2}}{x^2} + \frac{x^{1/3}}{x^2} + \frac{6}{x^2} \right) dx = \int \left(x^{-3/2} + x^{-5/3} + 6x^{-2} \right) dx \\ &= -2x^{-1/2} - \frac{3}{2}x^{-2/3} - 6x^{-1} + C \end{aligned}$$

$$(g) \int \frac{x^3}{x+1} dx.$$

Solution

Since the degree of the polynomial in the numerator is greater than that in the denominator, long division yields

$$\begin{array}{r} x^2 - x + 1 \\ x+1 \overline{) x^3} \\ \underline{-x^3 - x^2} \\ -x^2 \\ \underline{x^2 + x} \\ x \\ \underline{-x - 1} \\ -1 \end{array}$$

Therefore,

$$\int \frac{x^3}{x+1} dx = \int \left(x^2 - x + 1 - \frac{1}{x+1} \right) dx = \frac{x^3}{3} - \frac{x^2}{2} + x - \ln|x+1| + C$$

4.2 The u -substitution method

This technique requires that a new variable, say u , be introduced in the integrand to reduce the problem to a form in which the power rule of integration can be applied.

Example(s):

(a) Evaluate $\int (2x+1)^{1/3} dx$.

Solution

Let $u = 2x + 1$. Differentiating with respect to x yields $\frac{du}{dx} = 2 \Rightarrow dx = \frac{du}{2}$. Substituting into the given integral, we get

$$\int (2x+1)^{1/3} dx = \int u^{1/3} \frac{du}{2} = \frac{1}{2} \int u^{1/3} du = \frac{1}{2} \left[\frac{3}{4} u^{4/3} \right] + C = \frac{3}{8} (2x+1)^{4/3} + C$$

(b) Evaluate $\int \frac{x}{(1-x^2)^3} dx$.

Solution

Let $u = 1 - x^2$. Differentiating with respect to x yields $\frac{du}{dx} = -2x \Rightarrow dx = -\frac{du}{2x}$. Substituting into the given integral, we get

$$\int \frac{x}{(1-x^2)^3} dx = \int \frac{x}{u^3} \left(-\frac{du}{2x} \right) = -\frac{1}{2} \int u^{-3} du = -\frac{1}{2} \left[\frac{1}{-2} u^{-2} \right] + C = \frac{1}{4} (1-x^2)^{-2} + C$$

(c) Evaluate $\int 5x(1+3x)^5 dx$.

Solution

Let $u = 1 + 3x$. Differentiating with respect to x yields $\frac{du}{dx} = 3$. Therefore, $dx = \frac{du}{3}$ and $x = \frac{1}{3}(u-1)$. Substituting into the given integral, we get

$$\begin{aligned} \int 5x(1+3x)^5 dx &= \int \frac{5}{3} (u-1) u^5 \left(\frac{du}{3} \right) = \frac{5}{9} \int (u^6 - u^5) du = \frac{5}{9} \left[\frac{1}{7} u^7 - \frac{1}{6} u^6 \right] + C \\ &= \frac{5}{9} \left[\frac{1}{7} (1+3x)^7 - \frac{1}{6} (1+3x)^6 \right] + C \end{aligned}$$

Exercise:

1. Evaluate the following integrals

(a) $\int (7x-2)^3 dx$. [hint: put $u = 7x-2$, ans: $= \frac{1}{28} (7x-2)^4 + C$]

(b) $\int 3x\sqrt{1+2x^2} dx$. [hint: put $u = 1+2x^2$, ans: $= \frac{1}{2} (1+2x^2)^{3/2} + C$]

(c) $\int \frac{1}{(x+1)^2} dx$. [hint: put $u = x+1$, ans: $= \frac{-1}{x+1} + C$]

(d) $\int (3x+5)^{-2} dx$. [hint: put $u = 3x+5$, ans: $= -\frac{1}{3} (3x+5)^{-1} + C$]

- (e) $\int \frac{(\ln x)^2}{x} dx$. [hint: put $u = \ln x$, ans: $= \frac{1}{3}(\ln x)^3 + C$]
- (f) $\int x(3 - 5x^2)^4 dx$. [hint: put $u = 3 - 5x^2$, ans: $= -\frac{1}{50}(3 - 5x^2)^5 + C$]
- (g) $\int \frac{x+1}{\sqrt[5]{x^2+2x+7}} dx$. [hint: put $u = x^2 + 2x + 7$, ans: $= \frac{5}{8}(x^2 + 2x + 7)^{4/5} + C$]
- (h) $\int x(2+3x)^4 dx$. [hint: put $u = 2 + 3x$, ans: $= \frac{1}{3} \left(\frac{(2+3x)^6}{18} - \frac{2(2+3x)^5}{15} \right) + C$]
- (i) $\int x^2(1+4x^3)^3 dx$. [hint: put $u = 1 + 4x^3$, ans: $= \frac{1}{48}(1 + 4x^3)^4 + C$]
- (j) $\int \frac{1}{\sqrt{x}(1+\sqrt{x})^4} dx$. [hint: put $u = 1 + \sqrt{x}$, ans: $= -\frac{2}{3}(1 + \sqrt{x})^{-3} + C$]

2. If the marginal revenue function for output x is given by $MR = \frac{6}{(x+2)^2} + 5$, find the total revenue function and the demand equation. [hint: $R(0) = 0$, ans: revenue $R(x) = -\frac{6}{x+2} + 5x$, demand $p = \frac{R(x)}{x}$]

□ **The u -substitution based on the definition of integration as the reverse process of differentiation**

We consider the following forms of integrals.

4.2.1 Integrals of the form $\int f'(x)e^{f(x)} dx$

In this case, we let $u = f(x)$. Differentiating with respect to x yields $du = f'(x)dx$. Substituting yields

$$\int f'(x)e^{f(x)} dx = \int e^u du = e^u + c = e^{f(x)} + c,$$

where c is a constant of integration.

Example(s):

- (a) Evaluate $\int e^{3x} dx$

Solution

Let $u = 3x$. Differentiating with respect to x , we get $\frac{du}{dx} = 3 \Rightarrow dx = \frac{du}{3}$. Substituting yields,

$$\int e^{3x} dx = \int e^u \frac{du}{3} = \frac{1}{3} \int e^u du = \frac{1}{3} e^u + c = \frac{1}{3} e^{3x} + c$$

- (b) Evaluate $\int x^2 e^{2x^3-4} dx$

Solution

Let $u = 2x^3 - 4$. Differentiating with respect to x , we get $\frac{du}{dx} = 6x^2 \Rightarrow x^2 dx = \frac{du}{6}$. Substituting yields,

$$\int x^2 e^{2x^3-4} dx = \int e^u \frac{du}{6} = \frac{1}{6} \int e^u du = \frac{1}{6} e^u + c = \frac{1}{6} e^{2x^3-4} + c$$

Exercise:

(a) Evaluate $\int \cos(4x)e^{\sin 4x} dx$. [hint: let $u = \sin 4x$, ans: $= \frac{1}{4}e^{\sin 4x} + c$]

(b) Is it possible to evaluate $\int e^{x^2} dx$?

4.2.2 Integrals of the form $\int \frac{f'(x)}{f(x)} dx$

In this case, we let $u = f(x)$. Differentiating with respect to x yields $du = f'(x)dx$. Substituting yields

$$\int \frac{f'(x)}{f(x)} dx = \int \frac{1}{u} du = \ln |u| + c = \ln |f(x)| + c,$$

where c is a constant of integration.

Example(s):

(a) Evaluate $\int \frac{x-3}{x^2-6x+2} dx$.

Solution

Let $u = x^2 - 6x + 2$. Differentiating with respect to x yields $\frac{du}{dx} = 2x - 6 \Rightarrow dx = \frac{du}{2(x-3)}$.

Substituting into the given integral, we get

$$\int \frac{x-3}{x^2-6x+2} dx = \int \frac{(x-3)}{u} \cdot \frac{du}{2(x-3)} = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln |x^2 - 6x + 2| + C$$

(b) Evaluate $\int \tan x dx$.

Solution

Using the trigonometric identity $\tan x = \frac{\sin x}{\cos x}$, we have

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx$$

Let $u = \cos x$. Differentiating with respect to x yields $\frac{du}{dx} = -\sin x \Rightarrow dx = -\frac{du}{\sin x}$. Substituting into the given integral, we get

$$\int \frac{\sin x}{\cos x} dx = \int \frac{\sin x}{u} \left(-\frac{du}{\sin x} \right) = -\int \frac{1}{u} du = -\ln |u| + C = -\ln |\cos x| + C = \ln |\sec x| + C$$

Therefore,

$$\int \tan x dx = \ln |\sec x| + C$$

(c) Evaluate $\int \cot x dx$.

Solution

Using the trigonometric identity $\cot x = \frac{\cos x}{\sin x}$, we have

$$\int \cot x dx = \int \frac{\cos x}{\sin x} dx$$

Let $u = \sin x$. Differentiating with respect to x yields $\frac{du}{dx} = \cos x \Rightarrow dx = \frac{du}{\cos x}$. Substituting into the given integral, we get

$$\int \frac{\cos x}{\sin x} dx = \int \frac{\cos x}{u} \left(\frac{du}{\cos x} \right) = \int \frac{1}{u} du = \ln |u| + C = \ln |\sin x| + C$$

Therefore,

$$\int \cot x dx = \ln |\sin x| + C$$

(d) Evaluate $\int \sec x dx$.

Solution

Multiplying the numerator and denominator by $(\sec x + \tan x)$, we get

$$\int \sec x dx = \int \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$$

Let $u = \sec x + \tan x$. Differentiating with respect to x yields

$$\frac{du}{dx} = \sec x \tan x + \sec^2 x \Rightarrow dx = \frac{du}{\sec x \tan x + \sec^2 x}$$

Substituting into the given integral, we get

$$\begin{aligned} \int \sec x dx &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx = \int \frac{(\sec^2 x + \sec x \tan x)}{u} \cdot \frac{du}{(\sec x \tan x + \sec^2 x)} \\ &= \int \frac{1}{u} du = \ln |u| + C = \ln |\sec x + \tan x| + C \end{aligned}$$

Therefore,

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

(e) Evaluate $\int \operatorname{cosec} x dx$.

Solution

Multiplying the numerator and denominator by $(\operatorname{cosec} x + \cot x)$, we get

$$\int \operatorname{cosec} x dx = \int \frac{\operatorname{cosec} x (\operatorname{cosec} x + \cot x)}{\operatorname{cosec} x + \cot x} dx = \int \frac{\operatorname{cosec}^2 x + \operatorname{cosec} x \cot x}{\operatorname{cosec} x + \cot x} dx$$

Let $u = \operatorname{cosec} x + \cot x$. Differentiating with respect to x yields

$$\frac{du}{dx} = -\operatorname{cosec} x \cot x - \operatorname{cosec}^2 x \Rightarrow dx = -\frac{du}{(\operatorname{cosec} x \cot x + \operatorname{cosec}^2 x)}$$

Substituting into the given integral, we get

$$\begin{aligned} \int \frac{\operatorname{cosec}^2 x + \operatorname{cosec} x \cot x}{\operatorname{cosec} x + \cot x} dx &= \int \frac{\operatorname{cosec}^2 x + \operatorname{cosec} x \cot x}{u} \left(-\frac{du}{\operatorname{cosec} x \cot x + \operatorname{cosec}^2 x} \right) \\ &= -\int \frac{1}{u} du = -\ln |u| + C = -\ln |\operatorname{cosec} x + \cot x| + C \end{aligned}$$

Therefore,

$$\int \operatorname{cosec} x dx = -\ln |\operatorname{cosec} x + \cot x| + C$$

Exercise:

(a) Evaluate $\int \frac{x}{1-3x^2} dx$.

[hint: let $u = 1 - 3x^2$, ans: $= -\frac{1}{6} \ln |1 - 3x^2| + c$]

(b) Evaluate $\int \frac{4x^2}{x^3-7} dx$

- (c) Evaluate $\int \frac{e^{3x}}{2 + e^{3x}} dx$. [hint: let $u = 2 + e^{3x}$, ans: $= \frac{1}{3} \ln |2 + e^{3x}| + c$]
- (d) Evaluate $\int \frac{\cos 3x}{1 + \sin 3x} dx$. [hint: let $u = 1 + \sin 3x$, ans: $= \frac{1}{3} \ln |1 + \sin 3x| + c$]
- (e) Evaluate $\int \frac{\sec^2 2x}{1 + \tan 2x} dx$

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4.2.3 Integrals of the form $\int g'(x) \cdot f(g(x)) dx$

In this case, we let $u = g(x)$. Differentiating with respect to x yields $du = g'(x)dx$. Substituting yields

$$\int g'(x) \cdot f(g(x)) dx = \int f(u) du + c,$$

where c is a constant of integration.

Example(s):

- (a) Evaluate $\int x \sin(6 - 3x^2) dx$.

Solution

Let $u = 6 - 3x^2$. Differentiating with respect to x yields $\frac{du}{dx} = -6x \Rightarrow dx = -\frac{du}{6x}$. Substituting into the given integral, we get

$$\int x \sin(6 - 3x^2) dx = \int x \sin(u) \left(-\frac{du}{6x}\right) = -\frac{1}{6} \int \sin(u) du = \frac{1}{6} \cos(u) + C = \frac{1}{6} \cos(6 - 3x^2) + C$$

- (b) Evaluate $\int \frac{\cos(\sqrt{x})}{\sqrt{x}} dx$.

Solution

Let $u = \sqrt{x}$. Differentiating with respect to x yields $\frac{du}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow dx = 2\sqrt{x} du$. Substituting into the given integral, we get

$$\int \frac{\cos(\sqrt{x})}{\sqrt{x}} dx = \int \frac{\cos(u)}{\sqrt{x}} (2\sqrt{x} du) = 2 \int \cos(u) du = 2 \sin(u) + C = 2 \sin(\sqrt{x}) + C$$

- (c) Evaluate $\int \sec^2(7x - 1) dx$.

Solution

Let $u = 7x - 1$. Differentiating with respect to x yields $\frac{du}{dx} = 7 \Rightarrow dx = \frac{du}{7}$. Substituting into the given integral, we get

$$\int \sec^2(7x - 1) dx = \int \sec^2(u) \cdot \frac{du}{7} = \frac{1}{7} \int \sec^2(u) du = \frac{1}{7} \tan(u) + C = \frac{1}{7} \tan(7x - 1) + C$$

- (d) Evaluate $\int \operatorname{cosec}^2(4x) dx$.

Solution

Let $u = 4x$. Differentiating with respect to x yields $\frac{du}{dx} = 4 \Rightarrow dx = \frac{du}{4}$. Substituting into the given integral, we get

$$\int \operatorname{cosec}^2(4x)dx = \int \operatorname{cosec}^2(u) \cdot \frac{du}{4} = \frac{1}{4} \int \operatorname{cosec}^2(u)du = -\frac{1}{4} \cot(u) + C = -\frac{1}{4} \cot(4x) + C$$

(e) Evaluate $\int \tan(5x - 2)dx$.

Solution

Let $u = 5x - 2$. Differentiating with respect to x yields $\frac{du}{dx} = 5 \Rightarrow dx = \frac{du}{5}$. Substituting into the given integral, we get

$$\int \tan(5x - 2)dx = \int \tan(u) \cdot \frac{du}{5} = \frac{1}{5} \int \tan(u)du = \frac{1}{5} \ln |\sec u| + C = \frac{1}{5} \ln |\sec(5x - 2)| + C$$

(f) Evaluate $\int (x - 1) \cot(x^2 - 2x - 1)dx$.

Solution

Let $u = x^2 - 2x - 1$. Differentiating with respect to x yields $\frac{du}{dx} = 2x - 2 \Rightarrow dx = \frac{du}{2(x - 1)}$. Substituting into the given integral, we get

$$\begin{aligned} \int (x - 1) \cot(x^2 - 2x - 1)dx &= \int (x - 1) \cot(u) \cdot \frac{du}{2(x - 1)} = \frac{1}{2} \int \cot(u)du = \frac{1}{2} \ln |\sin(u)| + C \\ &= \frac{1}{2} \ln |\sin(x^2 - 2x - 1)| + C \end{aligned}$$

(g) Evaluate $\int 2x \sec(4x^2 + 5)dx$.

Solution

Let $u = 4x^2 + 5$. Differentiating with respect to x yields $\frac{du}{dx} = 8x \Rightarrow dx = \frac{du}{8x}$. Substituting into the given integral, we get

$$\begin{aligned} \int 2x \sec(4x^2 + 5)dx &= \int 2x \sec(u) \cdot \frac{du}{8x} = \frac{1}{4} \int \sec(u)du = \frac{1}{4} \ln |\sec(u) + \tan(u)| + C \\ &= \frac{1}{4} \ln |\sec(4x^2 + 5) + \tan(4x^2 + 5)| + C \end{aligned}$$

(h) Evaluate $\int \operatorname{cosec}(2x + 3)dx$.

Solution

Let $u = 2x + 3$. Differentiating with respect to x yields $\frac{du}{dx} = 2 \Rightarrow dx = \frac{du}{2}$. Substituting into the given integral, we get

$$\begin{aligned} \int \operatorname{cosec}(2x + 3)dx &= \int \operatorname{cosec}(u) \cdot \frac{du}{2} = \frac{1}{2} \int \operatorname{cosec}(u)du = -\frac{1}{2} \ln |\operatorname{cosec}(u) + \cot(u)| + C \\ &= -\frac{1}{2} \ln |\operatorname{cosec}(2x + 3) + \cot(2x + 3)| + C \end{aligned}$$

Exercise:

1. Evaluate the following

(a) $\int (x^2 + 1) \sin(x^3 + 3x - 5) dx$

(i) $\int \frac{\tan(\sqrt{x})}{\sqrt{x}} dx$

(b) $\int e^{2x} \sin(5 - e^{2x}) dx$

(j) $\int (x^2 - 1) \tan(3x^3 - 9x - 7) dx$

(c) $\int \cos(4x + 5) dx$

(k) $\int \frac{\cot(\sqrt{x})}{\sqrt{x}} dx$

(d) $\int e^{-3x} \cos(4 + e^{-3x}) dx$

(l) $\int \cot(2 - x) dx$

(e) $\int \frac{\sec^2(\sqrt{x})}{\sqrt{x}} dx$

(m) $\int e^{4x} \cot(1 - e^{4x}) dx$

(f) $\int \sin(2x) \sec^2(1 + \cos 2x) dx$

(n) $\int 2x^2 \sec(4\pi x^3 + 6) dx$

(g) $\int e^{3x} \operatorname{cosec}^2(1 + e^{3x}) dx$

(p) $\int (x + 2) \operatorname{cosec}(x^2 + 4x - 3) dx$

(h) $\int (x + 1) \operatorname{cosec}^2(2x^2 + 4x - 5) dx$

4.2.4 Integrals of the form $\int f(x) \cdot f'(x) dx$

In this case, we let $u = f(x)$. Differentiating with respect to x yields $du = f'(x) dx$. Substituting yields

$$\int f(x) \cdot f'(x) dx = \int u du = \frac{u^2}{2} + c = \frac{1}{2} [f(x)]^2 + c,$$

where c is a constant of integration.

Example(s):

(a) Evaluate $\int \sin x \cos x dx$

Solution

Let $u = \sin x \Rightarrow du = \cos x dx$. Therefore,

$$\int \sin x \cos x dx = \int u du = \frac{u^2}{2} + c = \frac{\sin^2 x}{2} + c$$

(b) Evaluate $\int \frac{\ln x}{x} dx$

Solution

$\int \frac{\ln x}{x} dx = \int \ln x \cdot \frac{1}{x} dx$. Let $u = \ln x \Rightarrow du = \frac{1}{x} dx$. Therefore,

$$\int \frac{\ln x}{x} dx = \int u du = \frac{u^2}{2} + c = \frac{\ln^2 x}{2} + c$$

(c) Evaluate $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$

Solution

$\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx = \int \sin^{-1} x \cdot \frac{1}{\sqrt{1-x^2}} dx$. Let $u = \sin^{-1} x \Rightarrow du = \frac{1}{\sqrt{1-x^2}} dx$. Therefore,

$$\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx = \int u du = \frac{u^2}{2} + c = \frac{[\sin^{-1} x]^2}{2} + c$$

Exercise:

1. Evaluate $\int 2 \tan x \sec^2 x dx$

4.3 Trigonometric and hyperbolic substitutions

Consider the following general forms of radicals:

4.3.1 Integrals of the form $\int \frac{1}{x^2 + A^2} dx$

In this case, we put $x = A \tan \theta$ and $dx = A \sec^2 \theta d\theta$.

□ **Justification:** The given integral can be rewritten as $\frac{1}{A^2} \int \frac{1}{\left\{1 + \frac{x^2}{A^2}\right\}} dx$. Comparing $1 + \frac{x^2}{A^2}$ with the trigonometric identity $1 + \tan^2 \theta = \sec^2 \theta$, we have

$$\frac{x^2}{A^2} = \tan^2 \theta \quad \Rightarrow \quad x = A \tan \theta$$

Differentiating with respect to θ yields $\frac{dx}{d\theta} = A \sec^2 \theta$. Making dx the subject yields $dx = A \sec^2 \theta d\theta$.

→ Note: we use the identity for tangent if the denominator of the integrand doesn't have a square root and the sign is positive.

Example(s):

(a) Evaluate $\int \frac{1}{x^2 + 4} dx$

Solution

Using trigonometric substitution, put $x = 2 \tan \theta \Rightarrow dx = 2 \sec^2 \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{x^2 + 4} dx &= \int \frac{2 \sec^2 \theta d\theta}{4 \tan^2 \theta + 4} = \int \frac{2 \sec^2 \theta d\theta}{4 \sec^2 \theta} = \frac{1}{2} \int d\theta = \frac{1}{2} \theta + c \\ &= \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c \end{aligned}$$

(b) Evaluate $\int \frac{1}{x^2 + 10x + 30} dx$

Solution

We first complete the square in the denominator to get

$$x^2 + 10x + 30 = x^2 + 10x + \left(\frac{10}{2}\right)^2 + 30 - \left(\frac{10}{2}\right)^2 = (x + 5)^2 + 5 \quad [\text{hint: } c = \left(\frac{b}{2}\right)^2]$$

Hence, $\int \frac{1}{x^2 + 10x + 30} dx = \int \frac{1}{(x + 5)^2 + 5} dx$. Using trigonometric substitution, put $x + 5 = \sqrt{5} \tan \theta \Rightarrow dx = \sqrt{5} \sec^2 \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{(x + 5)^2 + 5} dx &= \int \frac{\sqrt{5} \sec^2 \theta d\theta}{5 \tan^2 \theta + 5} = \int \frac{\sqrt{5} \sec^2 \theta d\theta}{5 \sec^2 \theta} = \frac{\sqrt{5}}{5} \int d\theta = \frac{1}{\sqrt{5}} \theta + c \\ &= \frac{1}{\sqrt{5}} \tan^{-1} \left(\frac{x + 5}{\sqrt{5}} \right) + c \end{aligned}$$

(c) Evaluate $\int \frac{1}{2x^2 + 12x + 32} dx$

Solution

We first complete the square in the denominator to get

$$2x^2 + 12x + 32 = 2(x^2 + 6x + 16) = 2 \left(x^2 + 6x + \left(\frac{6}{2}\right)^2 + 16 - \left(\frac{6}{2}\right)^2 \right) = 2 \left[(x + 3)^2 + 7 \right]$$

Hence, $\int \frac{1}{2x^2 + 12x + 32} dx = \frac{1}{2} \int \frac{1}{(x+3)^2 + 7} dx$. Using trigonometric substitution, put $x + 3 = \sqrt{7} \tan \theta \Rightarrow dx = \sqrt{7} \sec^2 \theta d\theta$. So,

$$\begin{aligned} \frac{1}{2} \int \frac{1}{(x+3)^2 + 7} dx &= \frac{1}{2} \int \frac{\sqrt{7} \sec^2 \theta d\theta}{7 \tan^2 \theta + 7} = \frac{1}{2} \int \frac{\sqrt{7} \sec^2 \theta d\theta}{7 \sec^2 \theta} = \frac{\sqrt{7}}{2 \times 7} \int d\theta = \frac{1}{2\sqrt{7}} \theta + c \\ &= \frac{1}{2\sqrt{7}} \tan^{-1} \left(\frac{x+3}{\sqrt{7}} \right) + c \end{aligned}$$

Exercise:

(a) Evaluate $\int \frac{1}{x^2 - 6x + 16} dx$. [ans: $I = \frac{1}{\sqrt{7}} \tan^{-1} \left(\frac{x-3}{\sqrt{7}} \right) + c$]

4.3.2 Integrals of the form $\int \frac{1}{\sqrt{A^2 - x^2}} dx$

In this case, we put $x = A \sin \theta$ and $dx = A \cos \theta d\theta$.

□ **Justification:** The given integral can be rewritten as $\frac{1}{A} \int \frac{1}{\sqrt{1 - \frac{x^2}{A^2}}} dx$. Comparing $1 - \frac{x^2}{A^2}$ with the trigonometric identity $1 - \sin^2 \theta = \cos^2 \theta$, we have

$$\frac{x^2}{A^2} = \sin^2 \theta \Rightarrow x = A \sin \theta$$

Differentiating with respect to θ yields $\frac{dx}{d\theta} = A \cos \theta$. Making dx the subject yields $dx = A \cos \theta d\theta$.

→ Note: we use the identity for sine (or cosine) if the denominator of the integrand has a square root and the sign is negative.

Example(s):

(a) Evaluate $\int \frac{1}{\sqrt{25 - x^2}} dx$

Solution

Using trigonometric substitution, put $x = 5 \sin \theta \Rightarrow dx = 5 \cos \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{\sqrt{25 - x^2}} dx &= \int \frac{5 \cos \theta d\theta}{\sqrt{25 - 25 \sin^2 \theta}} = \int \frac{5 \cos \theta d\theta}{\sqrt{25 \cos^2 \theta}} = \int \frac{5 \cos \theta d\theta}{5 \cos \theta} = \theta + c \\ &= \sin^{-1} \left(\frac{x}{5} \right) + c \end{aligned}$$

(b) Evaluate $\int \frac{1}{\sqrt{3 - 2x - x^2}} dx$

Solution

We first complete the square in the denominator to get

$$3 - 2x - x^2 = -(x^2 + 2x - 3) = - \left[x^2 + 2x + \left(\frac{2}{2} \right)^2 - 3 - \left(\frac{2}{2} \right)^2 \right] = - \left[(x+1)^2 - 4 \right] = 4 - (x+1)^2$$

Hence,

$$\int \frac{1}{\sqrt{3 - 2x - x^2}} dx = \int \frac{1}{\sqrt{4 - (x+1)^2}} dx$$

Using trigonometric substitution, put $x + 1 = 2 \sin \theta \Rightarrow dx = 2 \cos \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{\sqrt{4 - (x+1)^2}} dx &= \int \frac{2 \cos \theta d\theta}{\sqrt{4 - 4 \sin^2 \theta}} = \int \frac{2 \cos \theta d\theta}{\sqrt{4 \cos^2 \theta}} = \int d\theta = \theta + c \\ &= \sin^{-1} \left(\frac{x+1}{2} \right) + c \end{aligned}$$

Exercise:

(a) Evaluate $\int_0^1 \sqrt{\frac{x}{2-x}} dx$ [hint: put $x = 2 \sin^2 \theta$, ans: $I = \frac{\pi}{2} - 1$]

4.3.3 Integrals of the form $\int \frac{1}{\sqrt{x^2 + A^2}} dx$

In this case, we put $x = A \sinh \theta$ and $dx = A \cosh \theta d\theta$.

□ **Justification:** The given integral can be rewritten as $\frac{1}{A} \int \frac{1}{\sqrt{1 + \frac{x^2}{A^2}}} dx$. Comparing $1 + \frac{x^2}{A^2}$ with the hyperbolic identity $1 + \sinh^2 \theta = \cosh^2 \theta$, we have

$$\frac{x^2}{A^2} = \sinh^2 \theta \quad \Rightarrow \quad x = A \sinh \theta$$

Differentiating with respect to θ yields $\frac{dx}{d\theta} = A \cosh \theta$. Making dx the subject yields $dx = A \cosh \theta d\theta$.

→ Note: we use the identity for hyperbolic sine (or hyperbolic cosine) if the denominator of the integrand has a square root.

Example(s):

(a) Evaluate $\int \frac{1}{\sqrt{x^2 + 4}} dx$

Solution

Using hyperbolic substitution, put $x = 2 \sinh \theta \Rightarrow dx = 2 \cosh \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{\sqrt{x^2 + 4}} dx &= \int \frac{2 \cosh \theta d\theta}{\sqrt{4 \sinh^2 \theta + 4}} = \int \frac{2 \cosh \theta d\theta}{\sqrt{4 \cosh^2 \theta}} = \int \frac{2 \cosh \theta d\theta}{2 \cosh \theta} = \theta + c \\ &= \sinh^{-1} \left(\frac{x}{2} \right) + c \end{aligned}$$

Exercise:

(a) Evaluate $\int \frac{1}{\sqrt{x^2 + 5x + 12}} dx$ [ans: $I = \sinh^{-1} \left(\frac{x + \frac{5}{2}}{5.75} \right) + c$]

(b) Evaluate $\int \frac{1}{\sqrt{2x^2 + 8x + 15}} dx$

4.3.4 Integrals of the form $\int \frac{1}{\sqrt{x^2 - A^2}} dx$

In this case, we put $x = A \cosh \theta$ and $dx = A \sinh \theta d\theta$.

□ **Justification:** The given integral can be rewritten as $\frac{1}{A} \int \frac{1}{\sqrt{\frac{x^2}{A^2} - 1}} dx$. Comparing $\frac{x^2}{A^2} - 1$ with the hyperbolic identity $\cosh^2 \theta - 1 = \sinh^2 \theta$, we have

$$\frac{x^2}{A^2} = \cosh^2 \theta \quad \Rightarrow \quad x = A \cosh \theta$$

Differentiating with respect to θ yields $\frac{dx}{d\theta} = A \sinh \theta$. Making dx the subject yields $dx = A \sinh \theta d\theta$.

Example(s):

(a) Evaluate $\int \frac{1}{\sqrt{x^2 - 9}} dx$

Solution

Using hyperbolic substitution, put $x = 3 \cosh \theta \Rightarrow dx = 3 \sinh \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{\sqrt{x^2 - 9}} dx &= \int \frac{3 \sinh \theta d\theta}{\sqrt{9 \cosh^2 \theta - 9}} = \int \frac{3 \sinh \theta d\theta}{\sqrt{9 \sinh^2 \theta}} = \int \frac{3 \sinh \theta d\theta}{3 \sinh \theta} = \theta + c \\ &= \cosh^{-1} \left(\frac{x}{3} \right) + c \end{aligned}$$

Exercise:

(a) Evaluate $\int \frac{dx}{\sqrt{4x^2 - 9}}$. [hint: put $x = \frac{3}{2} \cosh \theta$, ans: $I = \ln \left[\frac{2x}{3} + \sqrt{\frac{4x^2}{9} - 1} \right] + c$]

(b) Evaluate $\int \frac{1}{\sqrt{x^2 + 6x + 1}} dx$ [ans: $I = \cosh^{-1} \left(\frac{x+3}{2\sqrt{2}} \right)$]

Exercise:

1. Evaluate the following integrals.

(a) $\int \frac{dx}{9x^2 - 25}$. [hint: put $x = \frac{5}{3} \coth \theta$, ans: $I = \frac{1}{30} \ln \left[\frac{3x - 5}{3x + 5} \right] + c$]

(b) $\int \sqrt{25 + x^2} dx$. [hint: put $x = \tan \theta$ then use integration by parts, ans:
 $I = \frac{5x}{2} \sec(\tan^{-1}(\frac{x}{5})) + \frac{25}{2} \ln[\sec(\tan^{-1}(\frac{x}{5})) + \frac{x}{5}] + c$]

2. Show that:

(a) $\int \sqrt{x^2 + 4} dx = \frac{1}{2} x \sqrt{x^2 + 4} + 2 \sinh^{-1}(x/2) + c$. [hint: put $x = 2 \sinh \theta$, recall:
 $\sinh 2\theta = 2 \sinh \theta \cosh \theta = 2 \sinh \theta \sqrt{1 + \sinh^2 \theta}$]

(b) $\int \frac{dx}{x\sqrt{1-x^2}} = -\operatorname{sech}^{-1} x + c$. [hint: put $x = \operatorname{sech} \theta$]

LECTURE 7

4.4 Integration of trigonometric functions**4.4.1 Integrals involving products of sines and cosines**

In this case, we rewrite the integrand using the following trigonometric identities.

$$\begin{aligned}\sin(A+B) &= \sin A \cos B + \cos A \sin B \\ \sin(A-B) &= \sin A \cos B - \cos A \sin B\end{aligned}$$

Adding yields

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

The restriction is that A is associated with sine and B is associated with cosine. Also,

$$\begin{aligned}\cos(A+B) &= \cos A \cos B - \sin A \sin B \\ \cos(A-B) &= \cos A \cos B + \sin A \sin B\end{aligned}$$

Adding yields

$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

Subtracting yields

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

Further, $\sin(-A) = -\sin A$ since sine is an odd function and $\cos(-A) = +\cos A$ since cosine is an even function.

Example(s):

- $\int \sin 4x \cos 2x dx = \frac{1}{2} \int (\sin 6x + \sin 2x) dx = -\frac{1}{12} \cos 6x - \frac{1}{4} \cos 2x + c$
- $\int \cos 3x \sin 5x dx = \frac{1}{2} \int [\sin 8x - \sin(-2x)] dx = \frac{1}{2} \int [\sin 8x + \sin 2x] dx = -\frac{1}{16} \cos 8x - \frac{1}{4} \cos 2x + c$
- $\int \cos 4x \cos 6x dx = \frac{1}{2} \int [\cos 10x + \cos(-2x)] dx = \frac{1}{2} \int [\cos 10x + \cos 2x] dx = \frac{1}{20} \sin 10x + \frac{1}{4} \sin 2x + c$
- $\int \sin 5x \sin x dx = \frac{1}{2} \int (\cos 4x - \cos 6x) dx = \frac{1}{8} \sin 4x - \frac{1}{12} \sin 6x + c$

Exercise:

- Evaluate $\int \sin 3x \cos 7x dx$. [ans: $I = \frac{1}{8} \cos 4x - \frac{1}{20} \cos 10x + c$]

4.4.2 Integrals involving powers of sines and cosines

□ **Case 1:** Even powers of sines and cosines

In this case, we use the following trigonometric identities to express the integrand in terms of cosines and sines of multiples of x . From

$$\cos^2 \theta - \sin^2 \theta = \cos 2\theta \text{ and } \cos^2 \theta + \sin^2 \theta = 1$$

Solving the above equations simultaneously yields the following identities.

- i) $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$.
 ii) $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$.

Example(s):

1. Evaluate the following integrals

$$(a) \int \sin^2 4x dx \quad (b) \int \cos^2 5x dx \quad (c) \int \cos^4 2x dx \quad (d) \int \cos 5x \sin^2 3x dx$$

Solution

$$(a) \int \sin^2 4x dx = \frac{1}{2} \int (1 - \cos 8x) dx = \frac{1}{2} \left(x - \frac{1}{8} \sin 8x \right) + c$$

$$(b) \int \cos^2 5x dx = \frac{1}{2} \int (1 + \cos 10x) dx = \frac{1}{2} \left(x + \frac{1}{10} \sin 10x \right) + c$$

(c)

$$\begin{aligned} \int \cos^4 2x dx &= \int (\cos^2 2x)^2 dx = \int \frac{(1 + \cos 4x)^2}{4} dx = \frac{1}{4} \int (1 + 2 \cos 4x + \cos^2 4x) dx \\ &= \frac{1}{4} \int \left(1 + 2 \cos 4x + \frac{1 + \cos 8x}{2} \right) dx = \frac{1}{4} \int \left(\frac{3}{2} + 2 \cos 4x + \frac{1}{2} \cos 8x \right) dx \\ &= \frac{1}{4} \left(\frac{3}{2} x + \frac{1}{2} \sin 4x + \frac{1}{16} \sin 8x \right) + c \end{aligned}$$

(d)

$$\begin{aligned} \int \cos 5x \sin^2 3x dx &= \int \cos 5x \left(\frac{1 - \cos 6x}{2} \right) dx = \frac{1}{2} \int (\cos 5x - \cos 5x \cos 6x) dx \\ &= \frac{1}{2} \int \left(\cos 5x - \frac{\cos 11x + \cos x}{2} \right) dx \\ &= \frac{1}{4} \int (2 \cos 5x - \cos 11x - \cos x) dx \\ &= \frac{1}{4} \left(\frac{2}{5} \sin 5x - \frac{1}{11} \sin 11x - \sin x \right) + c \end{aligned}$$

Exercise:

1. Evaluate the following integrals

$$\begin{array}{lll} (a) \int \sin^4 3x dx & (c) \int \cos^6 x dx & (e) \int \cos^6 x \sin 7x dx \\ (b) \int \sin^6 x dx & (d) \int \sin 3x \cos^2 2x dx & (f) \int \sin^2 x \cos 3x dx \end{array}$$

□ **Case 2:** Odd powers of sines and cosines

In this case, we use the trigonometric identity $\cos^2 \theta + \sin^2 \theta = 1$ after breaking the power into a product of even and odd powers of sines and/or cosines.

Example(s):

1. Evaluate the following integrals

$$(a) \int \sin^3 2x dx \quad (b) \int \cos^5 3x dx$$

Solution

(a)

$$\begin{aligned} \int \sin^3 2x dx &= \int \sin 2x \sin^2 2x dx = \int \sin 2x (1 - \cos^2 2x) dx \\ &= \int \sin 2x dx - \int \sin 2x \cos^2 2x dx \\ &= -\frac{1}{2} \cos 2x + \frac{1}{6} \cos^3 2x + c \end{aligned}$$

→ Note: for the second integral, put $u = \cos 2x \Rightarrow du = -2 \sin 2x dx$. Therefore,

$$\int \sin 2x \cos^2 2x dx = \int \sin 2x u^2 \left(-\frac{du}{2 \sin 2x} \right) = -\frac{1}{2} \int u^2 du = -\frac{1}{6} u^3 + c = -\frac{1}{6} \cos^3 2x + c$$

(b)

$$\begin{aligned} \int \cos^5 3x dx &= \int \cos 3x (\cos^2 3x)^2 dx = \int \cos 3x (1 - \sin^2 3x)^2 dx \\ &= \int \cos 3x (1 - 2 \sin^2 3x + \sin^4 3x) dx \\ &= \int \cos 3x dx - 2 \int \cos 3x \sin^2 3x dx + \int \cos 3x \sin^4 3x dx \\ &= \frac{1}{3} \sin 3x - \frac{2}{9} \sin^3 3x + \frac{1}{15} \sin^5 3x + c \end{aligned}$$

→ Note:

- For integrals involving powers of $\tan \theta$ and $\sec \theta$, use the trigonometric identity $1 + \tan^2 \theta = \sec^2 \theta$.
- For integrals involving powers of $\cot \theta$ and $\operatorname{cosec} \theta$, use the identity $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$.

Exercise:

- Evaluate $\int \tan^7 x dx$. [ans: $I = \frac{1}{6} \tan^6 x - \frac{1}{4} \tan^4 x + \frac{1}{2} \tan^2 x + \ln(\cos x) + c$]

Solution

□ **Method 1:** Using the trigonometric identity $1 + \tan^2 x = \sec^2 x$, we have

$$\int \tan^7 x dx = \int \tan^6 x \tan x dx = \int (\tan^2 x)^3 \tan x dx = \int (\sec^2 x - 1)^3 \tan x dx$$

Let $u = \sec x \Rightarrow du = \sec x \tan x dx$. Thus, $\tan x dx = \frac{du}{\sec x} = \frac{du}{u}$. Substituting yields

$$\begin{aligned} \int \tan^7 x dx &= \int \frac{(u^2 - 1)^3}{u} du = \int \frac{u^6 - 3u^4 + 3u^2 - 1}{u} du = \int \left(u^5 - 3u^3 + 3u - \frac{1}{u} \right) du \\ &= \frac{1}{6} u^6 - \frac{3}{4} u^4 + \frac{3}{2} u^2 - \ln(u) + c_1 \\ &= \frac{1}{6} \sec^6 x - \frac{3}{4} \sec^4 x + \frac{3}{2} \sec^2 x - \ln(\sec x) + c_1 \end{aligned}$$

□ **Method 2:** Using the trigonometric identity $1 + \tan^2 x = \sec^2 x$, we have

$$\begin{aligned} \tan^7 x &= \tan^2 x \tan^5 x = (\sec^2 x - 1) \tan^5 x = \tan^5 x \sec^2 x - \tan^5 x \\ \tan^5 x &= \tan^2 x \tan^3 x = (\sec^2 x - 1) \tan^3 x = \tan^3 x \sec^2 x - \tan^3 x \\ \tan^3 x &= \tan^2 x \tan x = (\sec^2 x - 1) \tan x = \tan x \sec^2 x - \tan x \end{aligned}$$

Back substitution yields

$$\begin{aligned}\tan^5 x &= \tan^3 x \sec^2 x - \tan x \sec^2 x + \tan x \\ \tan^7 x &= \tan^5 x \sec^2 x - \tan^3 x \sec^2 x + \tan x \sec^2 x - \tan x \\ &= (\tan^5 x - \tan^3 x + \tan x) \sec^2 x - \tan x\end{aligned}$$

Therefore,

$$\int \tan^7 x dx = \int (\tan^5 x - \tan^3 x + \tan x) \sec^2 x dx - \int \tan x dx$$

In the first integral, we let $u = \tan x \Rightarrow du = \sec^2 x dx$. Substituting yields

$$\begin{aligned}\int \tan^7 x dx &= \int (u^5 - u^3 + u) du - \int \frac{\sin x}{\cos x} dx \\ &= \frac{1}{6}u^6 - \frac{1}{4}u^4 + \frac{1}{2}u^2 + \ln(\cos x) + c \\ &= \frac{1}{6}\tan^6 x - \frac{1}{4}\tan^4 x + \frac{1}{2}\tan^2 x + \ln(\cos x) + c\end{aligned}$$

2. Evaluate the following integrals

$$(a) \int \sin^5 3x dx \quad (b) \int \cos^3 8x dx \quad (c) \int \sin^7 x dx \quad (d) \int \cos^7 x dx$$

4.4.3 Integrals involving rational functions with sines and cosines in the denominator

In this case, we use the t -method of substitution. We consider the following two cases.

Case 1: Integrals of the form $\int \frac{1}{a + b \sin^2 x + c \cos^2 x} dx$

In this case, we put $t = \tan x$ so that

$$\square \quad \frac{dt}{dx} = \sec^2 x \Rightarrow dx = \frac{dt}{\sec^2 x} = \frac{dt}{1 + \tan^2 x} \Rightarrow dx = \frac{dt}{1 + t^2}.$$

$$\square \quad \sin x = \frac{t}{\sqrt{1+t^2}} \text{ and } \cos x = \frac{1}{\sqrt{1+t^2}} \text{ (from SOHCAHTOA)}$$

Example(s):

$$1. \text{ Solve } \int \frac{1}{3 + \cos^2 x} dx$$

Solution

Put $t = \tan x \Rightarrow dx = \frac{dt}{1+t^2}$, and $\cos x = \frac{1}{\sqrt{1+t^2}}$. Hence,

$$\begin{aligned}\int \frac{1}{3 + \cos^2 x} dx &= \int \frac{1}{\left\{3 + \frac{1}{1+t^2}\right\} (1+t^2)} dt = \int \frac{(1+t^2)}{(4+3t^2)(1+t^2)} dt \\ &= \int \frac{1}{4+3t^2} dt. \text{ Again put } t = \frac{2}{\sqrt{3}} \tan \theta \Rightarrow dt = \frac{2}{\sqrt{3}} \sec^2 \theta d\theta \\ &= \frac{2}{\sqrt{3}} \int \frac{\sec^2 \theta}{[4+4\tan^2 \theta]} d\theta = \frac{2}{\sqrt{3}} \int \frac{\sec^2 \theta}{4\sec^2 \theta} d\theta = \frac{1}{2\sqrt{3}} \int d\theta \\ &= \frac{1}{2\sqrt{3}} \theta + c = \frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}t}{2} \right) + c = \frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3} \tan x}{2} \right) + c\end{aligned}$$

2. Solve $\int \frac{1}{2 \sin^2 x + 4 \cos^2 x} dx$

Solution

Put $t = \tan x \Rightarrow dx = \frac{dt}{1+t^2}$, $\sin x = \frac{t}{\sqrt{1+t^2}}$ and $\cos x = \frac{1}{\sqrt{1+t^2}}$. Hence,

$$\begin{aligned} \int \frac{1}{2 \sin^2 x + 4 \cos^2 x} dx &= \int \frac{1}{\left\{ 2 \cdot \frac{t^2}{1+t^2} + 4 \cdot \frac{1}{1+t^2} \right\} (1+t^2)} dt = \int \frac{(1+t^2)}{(2t^2+4)(1+t^2)} dt \\ &= \int \frac{1}{2t^2+4} dt. \text{ Again put } t = \sqrt{2} \tan \theta \Rightarrow dt = \sqrt{2} \sec^2 \theta d\theta \\ &= \int \frac{\sqrt{2} \sec^2 \theta}{[4 \tan^2 \theta + 4]} d\theta = \int \frac{\sqrt{2} \sec^2 \theta}{4 \sec^2 \theta} d\theta = \frac{\sqrt{2}}{4} \int d\theta \\ &= \frac{1}{2\sqrt{2}} \theta + c = \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{t}{\sqrt{2}} \right) + c = \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{\tan x}{\sqrt{2}} \right) + c \end{aligned}$$

Case 2: Integrals of the form $\int \frac{1}{a + b \sin x + c \cos x} dx$

In this case, we put $t = \tan \left(\frac{x}{2} \right)$ so that

$$\square \quad \frac{dt}{dx} = \frac{1}{2} \sec^2 \left(\frac{x}{2} \right) \Rightarrow dx = \frac{2dt}{\sec^2 \left(\frac{x}{2} \right)} = \frac{2dt}{1 + \tan^2 \left(\frac{x}{2} \right)} \Rightarrow dx = \frac{2dt}{1+t^2}$$

$$\square \quad \sin \left(\frac{x}{2} \right) = \frac{t}{\sqrt{1+t^2}} \text{ and } \cos \left(\frac{x}{2} \right) = \frac{1}{\sqrt{1+t^2}} \text{ (from SOHCAHTOA).}$$

$\square$ Therefore,

$$\sin x = 2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right) = 2 \cdot \frac{t}{\sqrt{1+t^2}} \cdot \frac{1}{\sqrt{1+t^2}} = \frac{2t}{1+t^2}$$

and

$$\cos x = \cos^2 \left(\frac{x}{2} \right) - \sin^2 \left(\frac{x}{2} \right) = \frac{1}{1+t^2} - \frac{t^2}{1+t^2} = \frac{1-t^2}{1+t^2}$$

Example(s):

1. Solve $\int \frac{1}{5 + 4 \cos x} dx$

Solution

Put $t = \tan \left(\frac{x}{2} \right) \Rightarrow dx = \frac{2dt}{1+t^2}$, $\cos \left(\frac{x}{2} \right) = \frac{1}{\sqrt{1+t^2}}$, and $\cos x = \frac{1-t^2}{1+t^2}$. Hence,

$$\begin{aligned} \int \frac{1}{5 + 4 \cos x} dx &= \int \frac{1}{\left\{ 5 + 4 \cdot \frac{1-t^2}{1+t^2} \right\} (1+t^2)} 2dt = 2 \int \frac{(1+t^2)}{(9+t^2)(1+t^2)} dt \\ &= 2 \int \frac{1}{9+t^2} dt. \text{ Again put } t = 3 \tan \theta \Rightarrow dt = 3 \sec^2 \theta d\theta \\ &= 2 \int \frac{3 \sec^2 \theta}{[9 + 9 \tan^2 \theta]} d\theta = 6 \int \frac{\sec^2 \theta}{9 \sec^2 \theta} d\theta = \frac{2}{3} \int d\theta \\ &= \frac{2}{3} \theta + c = \frac{2}{3} \tan^{-1} \left(\frac{t}{3} \right) + c = \frac{2}{3} \tan^{-1} \left(\frac{\tan \left(\frac{x}{2} \right)}{3} \right) + c \end{aligned}$$

2. Solve $\int \frac{1}{3 \sin x + 4 \cos x} dx$

Solution

Put $t = \tan\left(\frac{x}{2}\right) \Rightarrow dx = \frac{2dt}{1+t^2}$, $\cos\left(\frac{x}{2}\right) = \frac{1}{\sqrt{1+t^2}}$. Therefore, $\sin x = \frac{2t}{1+t^2}$ and $\cos x = \frac{1-t^2}{1+t^2}$. Hence,

$$\begin{aligned} \int \frac{1}{3 \sin x + 4 \cos x} dx &= \int \frac{1}{\left\{ 3 \cdot \frac{2t}{1+t^2} + 4 \cdot \frac{1-t^2}{1+t^2} \right\} (1+t^2)} 2dt = 2 \int \frac{1}{6t+4-4t^2} dt \\ &= -\frac{1}{2} \int \frac{1}{t^2 - \frac{3}{2}t - 1} dt = -\frac{1}{2} \int \frac{1}{\left(t - \frac{3}{4}\right)^2 - \frac{25}{16}} dt. \text{ (use partial fraction)} \\ &= -\frac{1}{2} \cdot \frac{1}{2 \cdot \frac{5}{4}} \ln \left(\frac{t - \frac{3}{4} - \frac{5}{4}}{t - \frac{3}{4} + \frac{5}{4}} \right) + c = -\frac{1}{5} \ln \left(\frac{t-2}{t+\frac{1}{2}} \right) + c \\ &= \frac{1}{5} \ln \left(\frac{2t+1}{2t-4} \right) + c = \frac{1}{5} \ln \left(\frac{2 \tan\left(\frac{x}{2}\right) + 1}{2 \tan\left(\frac{x}{2}\right) - 4} \right) + c \end{aligned}$$

Exercise:

1. Evaluate the following integrals

i) $\int \frac{1}{1 + \sin x - \cos x} dx$ [ans: $I = \ln \left(\frac{\tan\left(\frac{x}{2}\right)}{1 + \tan\left(\frac{x}{2}\right)} \right) + c$]

ii) $\int \sec x dx$ [ans: $I = \ln(\sec x + \tan x) + c$]

iii) $\int \frac{\cos x}{2 - \cos x} dx$ [hint: long div and t -subst., ans: $I = \frac{4}{\sqrt{3}} \tan^{-1} \left(\sqrt{3} \tan\left(\frac{x}{2}\right) \right) - x + c$]

iv) $\int \frac{1}{1 - 3 \sin x} dx$ [ans: $I = \frac{1}{\sqrt{8}} \ln \left(\frac{\tan\left(\frac{x}{2}\right) - 3 - \sqrt{8}}{\tan\left(\frac{x}{2}\right) - 3 + \sqrt{8}} \right) + c$]

v) $\int \frac{1}{2 + \cos x} dx$ [ans: $I = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\tan\left(\frac{x}{2}\right)}{\sqrt{3}} \right) + c$]

LECTURE 8

4.5 Integration by parts

The integration by parts formula, derived from the product rule of differentiation, is given as

$$\int u dv = uv - \int v du$$

Proof. Let u and v be functions of x . Differentiating the product uv with respect to x using the product rule of differentiation, we get

$$\frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Rearranging yields

$$u \frac{dv}{dx} = \frac{d(uv)}{dx} - v \frac{du}{dx}$$

Integrating both sides with respect to x yields

$$\int u \frac{dv}{dx} dx = \int \frac{d(uv)}{dx} dx - \int v \frac{du}{dx} dx$$

Simplifying yields $\int u dv = uv - \int v du$. □

→ Note: the parts formula is used when we have a product of two functions and either function is not the derivative of the other. We have to decide which function to take as u and which to link to dv . The following steps are necessary when using the parts formula.

□ To identify u , use the LIATEC rule. The remaining function(s) and dx form dv .

- **L** – Logarithmic function
- **I** – Inverse function
- **A** – Algebraic function
- **T** – Trigonometric function
- **E** – Exponential function
- **C** – Constant term

For example, to evaluate:

(a) $\int x^3 \ln x dx$, put $u = \ln x$ and $dv = x^3 dx$.

(b) $\int e^x \cos x dx$, put $u = \cos x$ and $dv = e^x dx$.

- Differentiate u with respect to x and make du the subject.
- Integrate dv with respect to x and disregard the constant of integration because a single constant will be added to the last step.
- Substitute the above results in the parts formula. Terminate the process if the last integrand can be integrated or repeat the process if the last integrand still contains product of functions or collect like terms if the last integrand is the same as the starting integrand.
- If the LIATEC rule fails, then u must be a group of functions that can be differentiated easily and dv must be a group of functions that can be integrated easily.

Example(s):

1. Evaluate $\int x^2 \ln x dx$

Solution

Put $u = \ln x$ and $dv = x^2 dx \Rightarrow du = \frac{1}{x} dx$ and $v = \frac{x^3}{3}$. Therefore,

$$\begin{aligned}\int x^2 \ln x dx &= uv - \int v du \\ &= \frac{x^3}{3} \ln x - \frac{1}{3} \int x^3 \cdot \frac{1}{x} dx = \frac{x^3}{3} \ln x - \frac{1}{3} \int x^2 dx \\ &= \frac{x^3}{3} \ln x - \frac{x^3}{9} + c\end{aligned}$$

2. Evaluate $\int x^2 \sin x dx$

Solution

Put $u = x^2$ and $dv = \sin x dx \Rightarrow du = 2x dx$ and $v = -\cos x$. Therefore,

$$\begin{aligned}\int x^2 \sin x dx &= uv - \int v du \\ &= -x^2 \cos x - \int (-\cos x) \cdot 2x dx = -x^2 \cos x + 2 \int x \cos x dx\end{aligned}$$

Since the last integrand still contains product of functions of x , we use another integration by parts. Put $u = x$ and $dv = \cos x dx \Rightarrow du = dx$ and $v = \sin x$. Therefore,

$$\begin{aligned}\int x \cos x dx &= uv - \int v du \\ &= x \sin x - \int \sin x dx = x \sin x + \cos x + c\end{aligned}$$

Hence, back substitution yields

$$\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + c_1$$

3. Evaluate $\int e^x \cos x dx$

Solution

Put $u = \cos x$ and $dv = e^x dx \Rightarrow du = -\sin x dx$ and $v = e^x$. Therefore,

$$\begin{aligned}\int e^x \cos x dx &= uv - \int v du \\ &= e^x \cos x - \int e^x \cdot (-\sin x dx) = e^x \cos x + \int e^x \sin x dx\end{aligned}$$

Since the last integrand still contains product of functions of x , we use another integration by parts. Put $u = \sin x$ and $dv = e^x dx \Rightarrow du = \cos x dx$ and $v = e^x$. Therefore,

$$\begin{aligned}\int e^x \sin x dx &= uv - \int v du \\ &= e^x \sin x - \int e^x \cos x dx\end{aligned}$$

Hence, back substitution yields

$$\int e^x \cos x dx = e^x \cos x + e^x \sin x - \int e^x \cos x dx$$

Since the last integrand is the same as the starting integrand, we collect like terms to get

$$2 \int e^x \cos x dx = e^x \cos x + e^x \sin x \Rightarrow \int e^x \cos x dx = \frac{1}{2} (e^x \cos x + e^x \sin x) + c$$

4. Evaluate $\int \ln x dx$

Solution

Put $u = \ln x$ and $dv = dx \Rightarrow du = \frac{1}{x}dx$ and $v = x$. Therefore,

$$\begin{aligned}\int \ln x dx &= uv - \int v du \\ &= x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int dx \\ &= x \ln x - x + c\end{aligned}$$

5. Evaluate $\int \sin^{-1} x dx$

Solution

Put $u = \sin^{-1} x$ and $dv = dx \Rightarrow du = \frac{1}{\sqrt{1-x^2}}dx$ and $v = x$. Therefore,

$$\begin{aligned}\int \sin^{-1} x dx &= uv - \int v du \\ &= x \sin^{-1} x - \int x \cdot \frac{1}{\sqrt{1-x^2}} dx = x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx, \text{ [put } x = \cos \theta] \\ &= x \sin^{-1} x + \sqrt{1-x^2} + c\end{aligned}$$

Alternatively: to evaluate $\int \frac{x}{\sqrt{1-x^2}} dx$, use the u -substitution method by putting $u = 1 - x^2$.

6. Evaluate $\int \sin^2 x dx$

Solution

$\int \sin^2 x dx = \int \sin x \cdot \sin x dx$. Put $u = \sin x$ and $dv = \sin x dx \Rightarrow du = \cos x dx$ and $v = -\cos x$. Therefore,

$$\begin{aligned}\int \sin^2 x dx &= uv - \int v du \\ &= -\sin x \cos x - \int (-\cos x) \cdot \cos x dx = -\sin x \cos x + \int \cos^2 x dx \\ &= -\sin x \cos x + \int (1 - \sin^2 x) dx = -\sin x \cos x + \int dx - \int \sin^2 x dx \\ \Rightarrow \int \sin^2 x dx + \int \sin^2 x dx &= -\sin x \cos x + \int dx \\ \Rightarrow 2 \int \sin^2 x dx &= -\sin x \cos x + x + c \\ \therefore \int \sin^2 x dx &= \frac{x}{2} - \frac{\sin x \cos x}{2} + c_1\end{aligned}$$

7. Evaluate $\int \sec^3 x dx$

Solution

$\int \sec^3 x dx = \int \sec x \cdot \sec^2 x dx$. Put $u = \sec x$ and $dv = \sec^2 x dx \Rightarrow du = \sec x \tan x dx$ and

$v = \tan x$. Therefore,

$$\begin{aligned}
 \int \sec^3 x dx &= uv - \int v du \\
 &= \sec x \tan x - \int \tan x \cdot \sec x \tan x dx = \sec x \tan x - \int \sec x \tan^2 x dx \\
 &= \sec x \tan x - \int \sec x [\sec^2 x - 1] dx = \sec x \tan x + \int \sec x dx - \int \sec^3 x dx \\
 \Rightarrow 2 \int \sec^3 x dx &= \sec x \tan x + \int \sec x dx \\
 &= \sec x \tan x + \ln[\sec x + \tan x] + c \\
 \therefore \int \sec^3 x dx &= \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln[\sec x + \tan x] + c_1
 \end{aligned}$$

Exercise:

1. Evaluate the following

$$(a) \int x^2 \sin x dx \quad [\text{ans: } I = -x^2 \cos x + 2x \sin x + 2 \cos x + c]$$

$$(b) \int x^3 e^{2x} dx \quad [\text{ans: } I = 8e^{2x} (4x^3 - 6x^2 + 6x - 3) + c]$$

$$(c) \int \frac{\ln x}{\sqrt{x}} dx \quad [\text{ans: } I = 2\sqrt{x} (\ln x - 2) + c]$$

$$(d) \int 2x^2 \cos^{-1} x dx. \quad [\text{ans: } I = \frac{2}{3} x^3 \cos^{-1} x - \frac{2}{3} \sqrt{1-x^2} + \frac{2}{9} (1-x^2)^{3/2} + c]$$

$$(e) \int x^4 \ln x dx. \quad [\text{ans: } I = \frac{x^5}{25} (5 \ln x - 1) + c]$$

$$(f) \int x^3 e^{x^2} dx \quad [\text{hint: put } u = x^2, \text{ ans: } I = \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2} + c]$$

$$2. \text{ Evaluate } \int_0^1 \frac{(\tan^{-1} x)^2}{1+x^2} dx \quad [\text{hint: put } u = (\tan^{-1} x)^2, dv = \frac{1}{1+x^2} dx, \text{ ans: } I = \frac{\pi^3}{192}, 7 \text{ Marks}]$$

$$3. \text{ Travis has the following information about the function } f(x): f(5) = 4, f(6) = 7 \text{ and } \int_5^6 e^{-7x} f(x) dx = 14. \text{ Evaluate } \int_5^6 e^{-7x} f'(x) dx. \quad [\text{ans: } I = 98]$$

4.6 Integration via partial fractions decomposition

The method of partial fractions decomposition is employed when integrating a rational function -i.e., a function of the form $\frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials in x . The rational function is expressed as a sum of simpler rational functions that can be integrated.

→ Note:

- ☐ Partial fraction decomposition is applied when the degree of $p(x)$ is less than that of $q(x)$ and $q(x)$ consists of linear and/or irreducible quadratic factors.
- ☐ If the degree of $p(x)$ is greater than or equal to the degree of $q(x)$, then we first perform the ordinary long division.

To decompose a rational function into its partial fractions, we consider the following four cases.

- ☐ **Case 1:** when $q(x)$ consists of non-repeated linear factors.

The partial fraction representation is

$$\frac{p(x)}{(x-a_1)(x-a_2)\cdots(x-a_n)} = \frac{A_1}{(x-a_1)} + \frac{A_2}{(x-a_2)} + \cdots + \frac{A_n}{(x-a_n)},$$

where $A_1, A_2, \dots, A_n$ are constants to be determined. For example,

$$\frac{2x+1}{(x+1)(x-1)} = \frac{A_1}{(x+1)} + \frac{A_2}{(x-1)} \Rightarrow A_1(x-1) + A_2(x+1) = 2x+1$$

Put $x = 1 \Rightarrow A_2 = \frac{3}{2}$. Put $x = -1 \Rightarrow A_1 = \frac{1}{2}$. Therefore,

$$\frac{2x+1}{(x+1)(x-1)} = \frac{1}{2} \left(\frac{1}{x+1} + \frac{3}{x-1} \right)$$

□ **Case 2:** when $q(x)$ consists of linear factors, at least one of which is repeated.

This is treated in the same manner as Case 1 but a repeated factor gives rise to partial fraction of the form

$$\frac{p(x)}{(x-a_1)(x-a_2)\cdots(x-a_n)(x-b)^k} = \frac{A_1}{(x-a_1)} + \frac{A_2}{(x-a_2)} + \cdots + \frac{A_n}{(x-a_n)} + \frac{B_1}{(x-b)} + \frac{B_2}{(x-b)^2} + \cdots + \frac{B_k}{(x-b)^k},$$

where $A_1, A_2, \dots, A_n; B_1, B_2, \dots, B_k$ are constants to be determined. E.g

$$\frac{3x+1}{(x-1)^2(x-2)} = \frac{A_1}{(x-1)} + \frac{A_2}{(x-1)^2} + \frac{A_3}{(x-2)}$$

□ **Case 3:** when $q(x)$ contains irreducible quadratic factors, but none is repeated.

The partial fraction representation is

$$\frac{p(x)}{(x-a_1)(x-a_2)\cdots(x-a_n)(bx^2+cx+d)} = \frac{A_1}{(x-a_1)} + \frac{A_2}{(x-a_2)} + \cdots + \frac{A_n}{(x-a_n)} + \frac{Bx+C}{(bx^2+cx+d)},$$

where $A_1, A_2, \dots, A_n; B, C$ are constants to be determined. For example,

$$\frac{x^2-1}{(x+2)(x^2+1)} = \frac{A_1}{(x+2)} + \frac{Bx+C}{(x^2+1)}$$

→ Note: a quadratic factor is a quadratic expression that cannot be factorized further (or an irreducible quadratic expression), e.g., (x^2+1) , (x^2+3x+5) , etc.

□ **Case 4:** when $q(x)$ contains at least one repeated irreducible quadratic factor.

The partial fraction representation is

$$\frac{p(x)}{(x-a_1)(x-a_2)\cdots(x-a_n)(bx^2+cx+d)^2} = \frac{A_1}{(x-a_1)} + \frac{A_2}{(x-a_2)} + \cdots + \frac{A_n}{(x-a_n)} + \frac{Bx+C}{(bx^2+cx+d)} + \frac{Dx+E}{(bx^2+cx+d)^2},$$

where $A_1, A_2, \dots, A_n; B, C, D, E$ are constants to be determined. E.g

$$\frac{x^3+1}{(x+1)(x^2+1)^2} = \frac{A_1}{(x+1)} + \frac{Bx+C}{(x^2+1)} + \frac{Dx+E}{(x^2+1)^2}$$

Example(s):

1. Evaluate $\int \frac{2x^3+x^2-6x+7}{x^2+x-6} dx$

Solution

We notice that the numerator has greater degree than the denominator. Thus, we divide the numerator by the denominator to get

$$\begin{array}{r} 2x-1 \\ x^2+x-6 \overline{) 2x^3+x^2-6x+7} \\ \underline{-2x^3-2x^2+12x} \\ -x^2+6x+7 \\ \underline{x^2+x-6} \\ 7x+1 \end{array}$$

Therefore, $\frac{2x^3 + x^2 - 6x + 7}{x^2 + x - 6} = 2x - 1 + \frac{7x + 1}{x^2 + x - 6}$.

Next: $x^2 + x - 6 = (x - 2)(x + 3)$. So, $\frac{7x + 1}{x^2 + x - 6} = \frac{7x + 1}{(x - 2)(x + 3)} = \frac{A}{(x - 2)} + \frac{B}{(x + 3)}$. Multiply both sides by $(x - 2)(x + 3)$ to get $A(x + 3) + B(x - 2) = 7x + 1$. Put $x = 2 \Rightarrow A = 3$. Put $x = -3 \Rightarrow B = 4$. Therefore, $\frac{7x + 1}{x^2 + x - 6} = \frac{3}{x - 2} + \frac{4}{x + 3}$. Now,

$$\begin{aligned} \int \frac{2x^3 + x^2 - 6x + 7}{x^2 + x - 6} dx &= \int \left(2x - 1 + \frac{3}{x - 2} + \frac{4}{x + 3} \right) dx \\ &= x^2 - x + 3 \ln(x - 2) + 4 \ln(x + 3) + c \end{aligned}$$

2. Evaluate $\int \frac{x^2}{x^3 - 3x^2 - 9x + 27} dx$

Solution

$x^3 - 3x^2 - 9x + 27 = (x - 3)^2(x + 3)$. So,

$$\frac{x^2}{x^3 - 3x^2 - 9x + 27} = \frac{x^2}{(x - 3)^2(x + 3)} = \frac{A}{(x - 3)} + \frac{B}{(x - 3)^2} + \frac{C}{(x + 3)}$$

Multiply both sides by $(x - 3)^2(x + 3)$ to get

$$A(x - 3)(x + 3) + B(x + 3) + C(x - 3)^2 = x^2$$

Put $x = 3 \Rightarrow B = \frac{3}{2}$. Put $x = -3 \Rightarrow C = \frac{1}{4}$. Put $x = 0 \Rightarrow -9A + 3B + 9C = 0 \Rightarrow A = \frac{3}{4}$.

$$\begin{aligned} \therefore \int \frac{x^2}{x^3 - 3x^2 - 9x + 27} dx &= \int \left(\frac{3}{4} \cdot \frac{1}{(x - 3)} + \frac{3}{2} \cdot \frac{1}{(x - 3)^2} + \frac{1}{4} \cdot \frac{1}{(x + 3)} \right) dx \\ &= \frac{3}{4} \int \frac{1}{x - 3} dx + \frac{3}{2} \int \frac{1}{(x - 3)^2} dx + \frac{1}{4} \int \frac{1}{x + 3} dx \\ &= \frac{3}{4} \ln|x - 3| - \frac{3}{2} \cdot \frac{1}{x - 3} + \frac{1}{4} \ln|x + 3| + c \end{aligned}$$

3. Evaluate $\int \frac{3x^3 - 18x^2 + 29x - 4}{(x + 1)(x - 2)^3} dx$

Solution

Let

$$\frac{3x^3 - 18x^2 + 29x - 4}{(x + 1)(x - 2)^3} = \frac{A}{(x + 1)} + \frac{B}{(x - 2)} + \frac{C}{(x - 2)^2} + \frac{D}{(x - 2)^3}$$

Multiply both sides by $(x + 1)(x - 2)^3$ to get

$$A(x - 2)^3 + B(x + 1)(x - 2)^2 + C(x + 1)(x - 2) + D(x + 1) = 3x^3 - 18x^2 + 29x - 4$$

Put $x = 2 \Rightarrow D = 2$. Put $x = -1 \Rightarrow A = 2$. Put $x = 0 \Rightarrow 2B - C = 5$ --- (i). Put $x = 1 \Rightarrow B - C = 4$ --- (ii). Solving (i) and (ii) simultaneously yields $B = 1, C = -3$.

$$\begin{aligned} \therefore \int \frac{3x^3 - 18x^2 + 29x - 4}{(x + 1)(x - 2)^3} dx &= \int \left(\frac{2}{(x + 1)} + \frac{1}{(x - 2)} - \frac{3}{(x - 2)^2} + \frac{2}{(x - 2)^3} \right) dx \\ &= \int \frac{2}{x + 1} dx + \int \frac{1}{x - 2} dx - \int \frac{3}{(x - 2)^2} dx + \int \frac{2}{(x - 2)^3} dx \\ &= 2 \ln|x + 1| + \ln|x - 2| + \frac{3}{x - 2} - \frac{2}{(x - 2)^2} + c \end{aligned}$$

4. Evaluate $\int \frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)} dx$

Solution

Let

$$\frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)} = \frac{A}{(2x - 1)} + \frac{Bx + C}{(x^2 + 4)}$$

Multiply both sides by $(2x - 1)(x^2 + 4)$ to get

$$A(x^2 + 4) + (Bx + C)(2x - 1) = x^2 - x - 21$$

Put $x = \frac{1}{2} \Rightarrow A = -5$. Put $x = 0 \Rightarrow C = 1$. Put $x = 1 \Rightarrow B = 3$.

$$\begin{aligned} \therefore \int \frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)} dx &= \int \left(\frac{-5}{2x - 1} + \frac{3x + 1}{x^2 + 4} \right) dx \\ &= -\frac{5}{2} \int \frac{2}{2x - 1} dx + \frac{3}{2} \int \frac{2x}{x^2 + 4} dx + \int \frac{1}{x^2 + 4} dx \\ &= -\frac{5}{2} \ln |2x - 1| + \frac{3}{2} \ln |x^2 + 4| + \int \frac{1}{x^2 + 4} dx \end{aligned}$$

Now, to evaluate $\int \frac{1}{x^2 + 4} dx$ put $x = 2 \tan \theta \Rightarrow dx = 2 \sec^2 \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{x^2 + 4} dx &= \int \frac{2 \sec^2 \theta d\theta}{4 \tan^2 \theta + 4} = \int \frac{2 \sec^2 \theta d\theta}{4 \sec^2 \theta} = \frac{1}{2} \int d\theta = \frac{1}{2} \theta + c \\ &= \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c \end{aligned}$$

Hence,

$$\int \frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)} dx = -\frac{5}{2} \ln |2x - 1| + \frac{3}{2} \ln |x^2 + 4| + \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c$$

Exercise:

1. Evaluate the following integrals

(a) $\int \frac{1}{3 - 2x - x^2} dx$. [ans: $I = \frac{1}{4} \ln |3 - 2x - x^2| + c$]

(b) $\int \frac{a^2}{(x + a)(x^2 + 2a^2)} dx$. [ans: $I = \frac{1}{3\sqrt{2}} \tan^{-1} \left(\frac{x}{a\sqrt{2}} \right) + \frac{1}{6} \ln \left| \frac{(x+a)^2}{x^2+a^2} \right| + c$]

(c) $\int \frac{1}{x^3 + 1} dx$. [hint: $x^3 + 1 = (x + 1)^3 - 3x(x + 1) = (x + 1)(x^2 - x + 1)$, ans:
 $I = \frac{1}{6} \ln \left| \frac{4(x+1)^2}{3(x^2-x+1)} \right| + \frac{\sqrt{3}}{3} \tan^{-1} \left(\frac{2x-1}{\sqrt{3}} \right) + c]$

(d) $\int \frac{4x^2 + 1}{x(2x - 1)^2} dx$. [ans: $I = \ln |x| - \frac{2}{2x - 1} + c$]

(e) $\int \frac{4x^2 - x + 12}{x(x^2 + 4)} dx$. [ans: $I = 3 \ln |x| - \frac{1}{2} \ln |x^2 + 4| - \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c$]

(f) $\int x^2 \ln(1 + x^2) dx$. [ans: $I = \frac{1}{9} (3x^3 \ln |1 + x^2| - 2x^3 + 6x - 6 \tan^{-1} x) + c$]

(g) $\int \frac{x^4 - 2x^3 + 3x^2 - x + 3}{x^3 - 2x^2 + 3x} dx$. [ans: $I = \frac{x^2}{2} + \ln \left| \frac{x}{\sqrt{x^2 - 2x + 3}} \right| + c$, 8 Marks]

(h) $\int \frac{x^4 - x^3 - x - 1}{x^3 - x^2} dx$. [ans: $I = \frac{x^2}{2} + \frac{1}{x} + 2 \ln \left| \frac{x - 1}{x} \right| + c$, 5 Marks]

(i) $\int \frac{1}{x(x^2 + 1)^2} dx$. [ans: $I = \ln \left| \frac{x}{\sqrt{x^2 + 1}} \right| + \frac{1}{2(x^2 + 1)} + c$]

(j) $\int \frac{10x - 2x^2}{(x-1)^2(x+3)} dx.$

[ans: $I = \ln \left| \frac{x-1}{(x+3)^3} \right| - \frac{2}{x-1} + c]$

(k) $\int \frac{16}{x^2 + 6x + 5} dx.$

[ans: $I = 4 \ln \left| \frac{x+1}{x+5} \right| + c]$

(l) $\int \frac{3x^2 - 7x - 2}{x^3 - x} dx.$

[ans: $I = 2 \ln |x| + 4 \ln |x+1| - 3 \ln |x-1| + c]$

(m) $\int \frac{1}{x(x^4 + 1)} dx.$

[ans: $I = \ln |x| - \frac{1}{4} \ln |x^4 + 1| + c]$

2. Evaluate $\int \frac{\sqrt{9-4x^2}}{x} dx.$

[hint: put $u^2 = 9 - 4x^2$ then use partial fractions. 5 Marks]

5 Applications of Integration in Ordinary Differential Equations

An ordinary differential equation (ODE) is an equation that contains derivatives of one or more dependent variables with respect to one independent variable. For example,

$$\frac{d^2y}{dx^2} + x \frac{dy}{dx} + 3y = 2xe^{x^2}$$

The order of an ODE is the order of the highest derivative involved in the equation. The degree of an ODE is the greatest power of the highest derivative involved in the equation. A first order ODE in two variables, say x and y , takes the general form

$$M(x, y)dx + N(x, y)dy = 0$$

5.1 Solution of first order ODE by the method of separation of variables

If in a first order ODE it is possible to collect all the functions of x and dx on one side and all the functions of y and dy on the other side, we say that the variables are separable. Thus, the above equation becomes

$$P(x)dx + Q(y)dy = 0$$

To get the solution, we integrate both sides to get

$$\int P(x)dx + \int Q(y)dy = C,$$

where C is a constant of integration.

Example(s):

1. Solve $\frac{dy}{dx} = \frac{2x}{y+1}$.

Solution

The given equation can be rewritten as $(y+1)dy = 2xdx$. Integrating yields

$$\int (y+1)dy = \int 2xdx \Rightarrow \frac{y^2}{2} + y = x^2 + c$$

2. Solve $\frac{dy}{dx} = (1+x)(1+y)$.

Solution

The given equation can be rewritten as $\frac{dy}{1+y} = (1+x)dx$. Integrating yields

$$\int \frac{dy}{1+y} = \int (1+x)dx \Rightarrow \ln|1+y| = x + \frac{x^2}{2} + c$$

3. Solve $\frac{dy}{dx} = e^{3x-2y} + x^2e^{-2y}$.

Solution

The given equation can be rewritten as

$$\frac{dy}{dx} = e^{3x}e^{-2y} + x^2e^{-2y} = e^{-2y}(e^{3x} + x^2) \Rightarrow e^{2y}dy = (e^{3x} + x^2)dx$$

Integrating yields

$$\int e^{2y}dy = \int (e^{3x} + x^2)dx \Rightarrow \frac{1}{2}e^{2y} = \frac{1}{3}e^{3x} + \frac{1}{3}x^3 + c$$

4. Solve $\frac{dy}{dx} = \frac{y^2 + xy^2}{x^2y - x^2}$.

Solution

The given equation can be rewritten as

$$\frac{dy}{dx} = \frac{y^2(1+x)}{x^2(y-1)} \Rightarrow \frac{y-1}{y^2} dy = \frac{x+1}{x^2} dx$$

Integrating yields

$$\int \frac{y-1}{y^2} dy = \int \frac{x+1}{x^2} dx \Rightarrow \int \left(\frac{1}{y} - \frac{1}{y^2} \right) dy = \int \left(\frac{1}{x} + \frac{1}{x^2} \right) dx \Rightarrow \ln y + \frac{1}{y} = \ln x - \frac{1}{x} + c$$

5. Solve $\frac{dy}{dx} = 2xy^2$, $y(0) = \frac{1}{2}$.

Solution

The given equation can be rewritten as $\frac{dy}{y^2} = 2x dx$. Integrating yields

$$\int \frac{dy}{y^2} = \int 2x dx \Rightarrow -\frac{1}{y} = x^2 + c \Rightarrow y = \frac{-1}{x^2 + c}$$

When $x = 0$, $y = \frac{1}{2}$. Hence, $\frac{1}{2} = \frac{-1}{0^2 + c} \Rightarrow c = -2$. Therefore, $y = \frac{-1}{x^2 - 2} = \frac{1}{2 - x^2}$.

Exercise:

1. Solve the following differential equations.

(a) $\frac{dy}{dx} = \frac{y^2 - 1}{x}$. [ans: $\frac{1}{2} \ln \left(\frac{y-1}{y+1} \right) = \ln x + c$]

(b) $xy \frac{dy}{dx} = \frac{x^2 + 1}{y + 1}$. [ans: $\frac{y^3}{3} + \frac{y^2}{2} = \frac{x^2}{2} + \ln x + c$]

(c) $x \frac{dy}{dx} = y + xy$. [ans: $y = Axe^x$]

(d) $y \tan x \frac{dy}{dx} = (4 + y^2) \sec^2 x$. [ans: $y^2 = A \tan^2 x - 4$]

(e) $\frac{dy}{dx} = \frac{y}{x}$. [ans: $y = Ax$]

(f) $\frac{\sin x}{1+y} \frac{dy}{dx} = \cos x$. [ans: $y = A \sin x - 1$]

(g) $2x dx + (1 - x^2) \cot y dy = 0$. [ans: $\sin y = c(1 - x^2)$]

(h) $\frac{dy}{dx} = \frac{xy}{x^2 + 1}$, $y(0) = 1$. [ans: $y^2 = x^2 + 1$]

6 Applications of Definite Integrals

6.1 Area between a curve and the x -axis

The area of the region bounded by the graph of a non-negative continuous function $y = f(x)$, the x -axis and the lines $x = a$ and $x = b$ is given by

$$A = \int_a^b f(x)dx = \int_a^b ydx$$

Example(s):

1. Find the area under the curve $y = x^2 - 4x + 5$ between $x = -1$ and $x = 2$.

Solution

$$y = x^2 - 4x + 5$$

x	-1	0	1	2
y	10	5	2	1

 (sketch the region)

$$A = \int_a^b ydx = \int_{-1}^2 (x^2 - 4x + 5)dx = \left[\frac{x^3}{3} - 2x^2 + 5x \right]_{-1}^2 = 12 \text{ square units}$$

2. Find the area between the curves $y = 1 - x^2$, $x = 2$ and $x = 5$.

Solution

$$y = 1 - x^2$$

x	2	3	4	5
y	-3	-8	-15	-24

 (sketch the region)

$$A = \int_a^b ydx = \int_2^5 (1 - x^2)dx = \left[x - \frac{x^3}{3} \right]_2^5 = -36 \quad \therefore A = 36 \text{ square units}$$

6.2 Area between two curves

Consider a region bounded by the curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$, where $f(x) \geq g(x)$ in the region $[a, b]$. Then, the area between the two curves from $x = a$ to $x = b$ is given by

$$A = \int_a^b [f(x) - g(x)] dx$$

Example(s):

1. Find the area between the curves $y = x^2$ and $y = x^2 + 2x + 3$ from $x = 1$ to $x = \frac{5}{2}$.

Solution

(sketch the region)

$$A = \int_1^{5/2} [(x^2 + 2x + 3) - x^2] dx = \int_1^{5/2} [2x + 3] dx = [x^2 + 3x]_1^{5/2} = 9\frac{3}{4} \text{ square units}$$

2. Find the area enclosed between the curves $y = 4 + 5x - x^2$ and $y = 4 - x$.

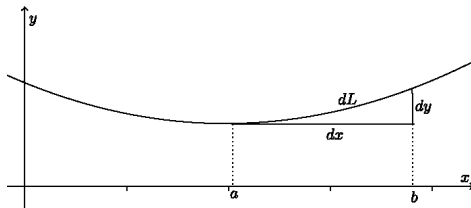
Solution

(sketch the region). The curves meet when $4 + 5x - x^2 = 4 - x \Rightarrow x = 0$ or $x = 6$.

$$A = \int_0^6 [(4 + 5x - x^2) - (4 - x)] dx = \int_0^6 [6x - x^2] dx = \left[3x^2 - \frac{x^3}{3} \right]_0^6 = 36 \text{ square units}$$

6.3 Arc length

Consider a curve defined by $y = f(x)$ for $x \in [a, b]$.



Let dx and dy be the small change in x and y , respectively. Then by Pythagoras theorem, the small change in its length L is given by

$$(dL)^2 = (dx)^2 + (dy)^2 \Rightarrow dL = \sqrt{(dx)^2 + (dy)^2}$$

Therefore, $dL = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$. Integrating from $x = a$ to $x = b$ yields

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx,$$

which is the length L of the arc joining the points $(a, f(a))$ and $(b, f(b))$ along the curve $y = f(x)$.

Alternatively, if the curve is given in parametric form, i.e. if we know $x(t)$ and $y(t)$, for $t \in [t_1, t_2]$, then

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Example(s):

1. Find the length of the curve $y = \frac{x^3}{6} + \frac{1}{2x}$ from $x = 1$ to $x = 2$.

Solution

$\frac{dy}{dx} = \frac{x^2}{2} - \frac{1}{2x^2}$. Hence, the arc length is

$$\begin{aligned} L &= \int_1^2 \sqrt{1 + \left[\frac{dy}{dx}\right]^2} dx = \int_1^2 \sqrt{1 + \frac{1}{4} \left[x^2 - \frac{1}{x^2}\right]^2} dx = \int_1^2 \sqrt{1 + \frac{1}{4} \left[x^4 - 2 + \frac{1}{x^4}\right]} dx \\ &= \frac{1}{2} \int_1^2 \sqrt{4 + \left[x^4 - 2 + \frac{1}{x^4}\right]} dx = \frac{1}{2} \int_1^2 \sqrt{x^4 + 2 + \frac{1}{x^4}} dx = \frac{1}{2} \int_1^2 \sqrt{\left(x^2 + \frac{1}{x^2}\right)^2} dx \\ &= \frac{1}{2} \int_1^2 \left(x^2 + \frac{1}{x^2}\right) dx = \frac{1}{2} \left[\frac{x^3}{3} - \frac{1}{x}\right]_1^2 = \frac{17}{12} \text{ units} \end{aligned}$$

2. Find the length of the curve $y = x^{3/2}$ between the points (1,1) and (4,8).

Solution

$\frac{dy}{dx} = \frac{3}{2}x^{1/2}$. Hence, the arc length is

$$\begin{aligned} L &= \int_1^4 \sqrt{1 + \left[\frac{dy}{dx}\right]^2} dx = \int_1^4 \sqrt{1 + \frac{9}{4}x} dx = \frac{1}{2} \int_1^4 \sqrt{4 + 9x} dx, \text{ (put } u^2 = 4 + 9x \Rightarrow 2udu = 9dx) \\ &= \frac{1}{2} \int_{\sqrt{13}}^{\sqrt{40}} u \cdot \frac{2udu}{9} = \frac{1}{9} \int_{\sqrt{13}}^{\sqrt{40}} u^2 du = \frac{1}{9} \left[\frac{u^3}{3} \right]_{\sqrt{13}}^{\sqrt{40}} = \frac{1}{27} [u^3]_{\sqrt{13}}^{\sqrt{40}} = \frac{1}{27} [80\sqrt{10} - 13\sqrt{13}] \text{ units} \end{aligned}$$

3. Find the length of the curve $x = 2\cos^3 \theta, y = 2\sin^3 \theta$ between the points corresponding to $\theta = 0$ and $\theta = \frac{\pi}{2}$.

Solution

$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{6\sin^2 \theta \cos \theta}{-6\cos^2 \theta \sin \theta} = -\tan \theta$. $dx = -6\cos^2 \theta \sin \theta d\theta$. Hence, the arc length is

$$\begin{aligned} L &= \int_0^{\pi/2} \sqrt{1 + \left[\frac{dy}{dx}\right]^2} \sqrt{(dx)^2} = \int_0^{\pi/2} \sqrt{1 + \tan^2 \theta} (6\cos^2 \theta \sin \theta d\theta) = 6 \int_0^{\pi/2} \sec \theta \cos^2 \theta \sin \theta d\theta \\ &= 6 \int_0^{\pi/2} \cos \theta \sin \theta d\theta = 3 \int_0^{\pi/2} 2\cos \theta \sin \theta d\theta = 3 \int_0^{\pi/2} \sin 2\theta d\theta = 3 \left[-\frac{\cos 2\theta}{2} \right]_0^{\pi/2} \\ &= -3 \left[\frac{\cos 2\theta}{2} \right]_0^{\pi/2} = -3 \left[\frac{-1}{2} - \frac{1}{2} \right] = 3 \text{ units} \end{aligned}$$

Exercise:

1. Prove that the length of the segment of the curve $y = x^{3/2}$ from $(0,0)$ to $(4,8)$ is $\frac{8}{27}[10\sqrt{10} - 1]$.
2. Find the length of the curve $x = 5(2t - \sin 2t), y = 10\sin^2 t$ between $t = 0$ to $t = \pi$. [ans: $L = 40$ units]
3. Find the length of the curve $y = \frac{x^3}{6} - \frac{1}{10x}$ from $x = 1$ to $x = 2$. [ans: $L = 2.67625$ units]
4. Find the length of the arc of the curve $(y + 1)^2 = 4x^3$ between $x = 1$ and $x = 2$. [ans: $L = 3.79232$ units]

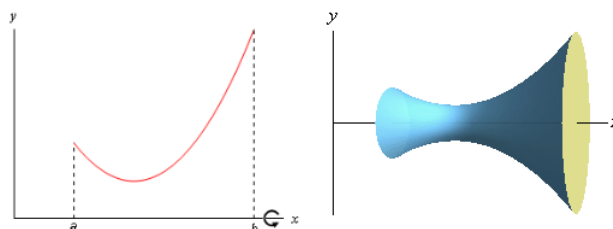
LECTURE 9

6.4 Area of surface of revolution

A surface of revolution is formed when a curve is rotated about a line.

□ Case 1: **Rotation about the x-axis.**

Consider rotating the curve $y = f(x), a \leq x \leq b$, about the x -axis, where f is positive and has a continuous derivative. E.g.



Then, area of the surface formed is defined by

$$dA = 2\pi y dL, \text{ where } dL = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \Rightarrow A = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

□ **Case 2: Rotation about the y-axis.**

Consider rotating the curve $x = h(y)$, $c \leq y \leq d$, about the y -axis, where h is positive and has a continuous derivative. Then, area of the surface formed is defined by

$$dA = 2\pi x dL, \text{ where } dL = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy \Rightarrow A = 2\pi \int_c^d x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

Example(s):

- Find the area of the surface obtained by rotating the arc $y = \sqrt{4 - x^2}$ about the x -axis given that $-1 \leq x \leq 1$.

Solution

We have

$$\frac{dy}{dx} = \frac{1}{2}(4 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{4 - x^2}}$$

So, the surface area is

$$\begin{aligned} A &= 2\pi \int_{-1}^1 y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 2\pi \int_{-1}^1 \sqrt{4 - x^2} \sqrt{1 + \frac{x^2}{4 - x^2}} dx \\ &= 2\pi \int_{-1}^1 \sqrt{4 - x^2} \frac{2}{\sqrt{4 - x^2}} dx = 4\pi \int_{-1}^1 dx = 4\pi(2) = 8\pi \end{aligned}$$

- The arc of the parabola $y = x^2$ from (1,1) to (2,4) is rotated about the y -axis. Find the area of the resulting surface.
[ans: $A = \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5})$]

Solution

We have

$$x = \sqrt{y} \text{ and } \frac{dx}{dy} = \frac{1}{2\sqrt{y}}$$

So, the surface area is

$$\begin{aligned} A &= 2\pi \int_1^4 x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy \\ &= 2\pi \int_1^4 \sqrt{y} \sqrt{1 + \frac{1}{4y}} dy = \pi \int_1^4 \sqrt{4y + 1} dy, \quad \text{put } u = 4y + 1 \Rightarrow du = 4dy \\ &= \frac{\pi}{4} \int_5^{17} \sqrt{u} du = \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5}) \end{aligned}$$

- Show that the area of the surface generated by rotating the curve $y = e^x$, $0 \leq x \leq 1$, about the x -axis is given by

$$S = \pi \left[e\sqrt{1 + e^2} + \ln|e + \sqrt{1 + e^2}| - \sqrt{2} - \ln|\sqrt{2} + 1| \right]$$

Solution

We have

$$y = e^x \text{ and } \frac{dy}{dx} = e^x$$

So, the surface area is

$$\begin{aligned} A &= 2\pi \int_0^1 y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dx = 2\pi \int_0^1 e^x \sqrt{1 + e^{2x}} dx, \text{ put } u = e^x \Rightarrow du = e^x dx \\ &= 2\pi \int_1^e \sqrt{1 + u^2} du, \text{ put } u = \tan \theta \Rightarrow du = \sec^2 \theta d\theta \\ &= 2\pi \int_{\pi/4}^{\tan^{-1} e} \sec^3 \theta d\theta, \text{ (since } \tan^{-1}(1) = \frac{\pi}{4}) \\ &= 2\pi \cdot \frac{1}{2} \left[\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta| \right]_{\pi/4}^{\tan^{-1} e} \\ &= \pi \left[\sec \tan^{-1}(e) \tan \tan^{-1}(e) + \ln |\sec \tan^{-1}(e) + \tan \tan^{-1}(e)| \right. \\ &\quad \left. - \sec(\pi/4) \tan(\pi/4) - \ln |\sec(\pi/4) + \tan(\pi/4)| \right] \\ &= \pi \left[e \sec \tan^{-1}(e) + \ln |\sec \tan^{-1}(e) + e| - \sqrt{2} - \ln |\sqrt{2} + 1| \right]. \text{ Recall that } \sec^2 \alpha = 1 + \tan^2 \alpha \\ &= \pi \left[e \sqrt{1 + \tan^2(\tan^{-1} e)} + \ln |\sqrt{1 + \tan^2(\tan^{-1} e)} + e| - \sqrt{2} - \ln |\sqrt{2} + 1| \right] \\ &= \pi \left[e \sqrt{1 + e^2} + \ln |e + \sqrt{1 + e^2}| - \sqrt{2} - \ln |\sqrt{2} + 1| \right] \end{aligned}$$

6.5 Volumes of solids of revolution

6.5.1 Disk method

If the axis of revolution is the boundary of the plane region and the cross sections are taken perpendicular to the axis of revolution, then by disk method, the volume (V) of the solid generated by revolving the region bounded by $y = f(x)$ and the x-axis on the interval $[a, b]$ about the x-axis is given by

$$dV = \pi y^2 dx \Rightarrow V = \pi \int_a^b y^2 dx$$

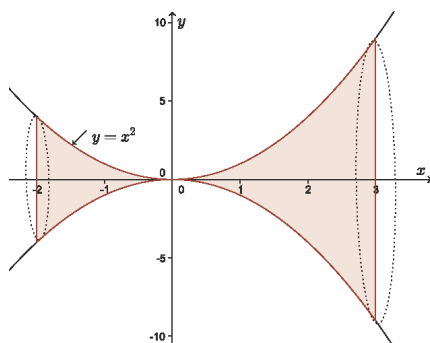
If the region bounded by $x = f(y)$ and the y-axis on the interval $[c, d]$ is revolved about the y-axis, then its volume (V) is given by

$$V = \pi \int_c^d x^2 dy$$

Example(s):

- (a) Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and the x-axis on $[-2, 3]$ about the x-axis.

Solution



Since the x-axis is a boundary of the region, we use the disk method. The volume of the solid generated is

$$\begin{aligned} V &= \pi \int_{-2}^3 y^2 dx = \pi \int_{-2}^3 x^4 dx \\ &= \pi \left[\frac{x^5}{5} \right]_{-2}^3 = \pi \left[\frac{243}{5} - \left(-\frac{32}{5} \right) \right] = 55\pi \end{aligned}$$

6.5.2 Washer method

If the axis of revolution is not a boundary of the plane region and the cross sections are taken perpendicular to the axis of revolution, then by washer¹ method, the volume (V) of the solid generated by revolving the region bounded by $y = f(x)$ and $y = g(x)$ on the interval $[a, b]$, where $f(x) \geq g(x)$, about the x-axis is given by

$$V = \pi \int_a^b \left([f(x)]^2 - [g(x)]^2 \right) dx$$

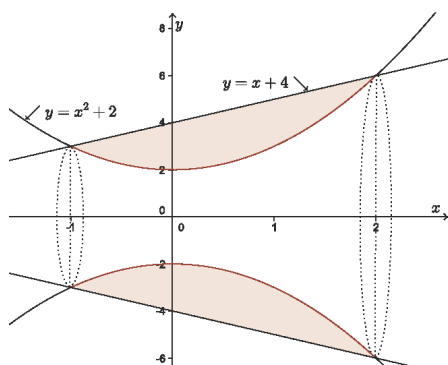
If the region bounded by $x = f(y)$ and $x = g(y)$ on the interval $[c, d]$, where $f(y) \geq g(y)$ is revolved about the y-axis, then its volume (V) is given by

$$V = \pi \int_c^d \left([f(y)]^2 - [g(y)]^2 \right) dy$$

Example(s):

- (a) Find the volume of the solid generated by revolving the region bounded by $y = x^2 + 2$ and $y = x + 4$ about the x-axis.

Solution



The curve $y = x^2 + 2$ and the line $y = x + 4$ intersect at $x = -1$ and $x = 2$. Since the x-axis is not a boundary of the region, we use the washer method. The volume of the solid generated is given by

$$\begin{aligned} V &= \pi \int_{-1}^2 \left([x + 4]^2 - [x^2 + 2]^2 \right) dx = \pi \int_{-1}^2 \left(x^2 + 8x + 16 - x^4 - 4x^2 - 4 \right) dx \\ &= \pi \int_{-1}^2 \left(-x^4 - 3x^2 + 8x + 12 \right) dx \\ &= \pi \left[-\frac{x^5}{5} - x^3 + 4x^2 + 12x \right]_{-1}^2 = \pi \left[\frac{128}{5} - \left(-\frac{34}{5} \right) \right] = \frac{162\pi}{5} \end{aligned}$$

¹Think of washer as a disk with a hole in it -i.e. centre is empty

6.5.3 Cylindrical shell method

If the cross sections of the solid generated are taken parallel to the axis of revolution, then by cylindrical shell method, the volume (V) of the solid generated by revolving the region bounded by $y = f(x)$ and the x-axis on the interval $[a, b]$, where $f(x) \geq 0$, about the y-axis is given by

$$V = 2\pi \int_a^b xy dx$$

If the region bounded by $x = f(y)$ and the y-axis on the interval $[c, d]$, where $f(y) \geq 0$, is revolved about the x-axis, then its volume (V) is given by

$$V = 2\pi \int_c^d yx dy$$

Example(s):

- (a) Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and the x-axis on the interval $[1, 3]$ about the y-axis.

Solution

The radius of the shell is x and the height of the shell is $y = x^2$. The volume of the solid generated is

$$\begin{aligned} V &= 2\pi \int_1^3 x \cdot x^2 dx = 2\pi \int_1^3 x^3 dx \\ &= 2\pi \left[\frac{x^4}{4} \right]_1^3 = 40\pi \end{aligned}$$

LECTURE 10

7 Numerical Integration

Many functions such as $\sqrt{x^3+1}$ or e^{x^2} do not have antiderivatives that are easily found. For such functions, the fundamental theorem of calculus is not useful and we can only approximate definite integrals involving these functions. Thus, we need to approximate the area of the region bounded by the curve $y = f(x)$, the lines $x = a$ and $x = b$, and the x-axis, i.e.,

$$A = \int_a^b f(x) dx$$

7.1 Mid-ordinate rule

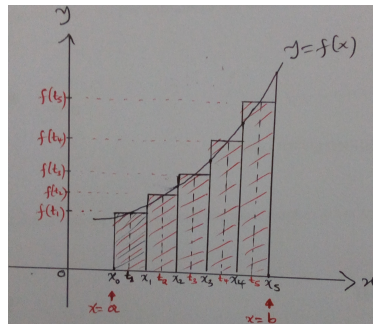
One way to approximate is to use sums of the form

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(t_i) \Delta x, \text{ where } \Delta x = \frac{b-a}{n}$$

and t_i is the mid-point of x_{i-1} and x_i i.e $t_i = \frac{x_{i-1} + x_i}{2}$. Letting $\Delta x = h$ yields

$$\int_a^b f(x) dx \approx h \sum_{i=1}^n f(t_i) = h [\text{sum of mid-ordinates}]$$

The above sum involves approximating slices of the region R bounded by the graph of $y = f(x)$ with rectangles of length $f(t_i)$ and width h . This is called the mid-ordinate rule.



Example(s):

1. Approximate $\int_1^4 \frac{1}{x} dx$ using the mid-ordinate rule with 7 ordinates.

Solution

$$f(x) = \frac{1}{x}, a = 1, b = 4, n = 6, h = \frac{b-a}{n} = \frac{4-1}{6} = \frac{1}{2}$$

	t_1	t_2	t_3	t_4	t_5	t_6
t	$\frac{5}{4}$	$\frac{7}{4}$	$\frac{9}{4}$	$\frac{11}{4}$	$\frac{13}{4}$	$\frac{15}{4}$
$f(t)$	$\frac{4}{5}$	$\frac{4}{7}$	$\frac{4}{9}$	$\frac{4}{11}$	$\frac{4}{13}$	$\frac{4}{15}$
	f_1	f_2	f_3	f_4	f_5	f_6

Therefore,

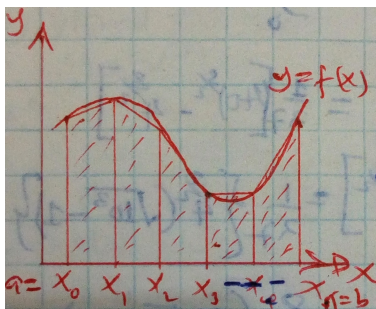
$$\begin{aligned} \int_1^4 \frac{1}{x} dx &\approx h [f_1 + f_2 + f_3 + f_4 + f_5 + f_6] \\ &= \frac{1}{2} \left[\frac{4}{5} + \frac{4}{7} + \frac{4}{9} + \frac{4}{11} + \frac{4}{13} + \frac{4}{15} \right] = 1.3769 \text{ sq. units} \end{aligned}$$

7.2 Trapezoidal rule

An approximating rectangle often doesn't fit the graph $y = f(x)$ very well. There may be large regions under the graph of $f(x)$ that are not included in the rectangle and vice versa. We can try to improve this rectangular estimate by drawing a line from $(x_{i-1}, f(x_{i-1}))$ to $(x_i, f(x_i))$, $i = 1, 2, \dots, n$ and sum the areas of the resulting trapezia. Thus, the area is given by

$$\int_a^b f(x) dx \approx \frac{h}{2} \sum_{i=1}^n [f(x_{i-1}) + f(x_i)] = \frac{h}{2} [f_0 + 2(f_1 + f_2 + \dots + f_{n-2} + f_{n-1}) + f_n]$$

This is called the trapezoidal rule.



→ Note: The bound for the truncation error (E) in using the trapezoidal rule is given as

$$|E| \leq \frac{(b-a)}{12} h^2 \max_{a \leq x \leq b} |f''(x)|$$

Example(s):

1. i) Approximate $\int_1^4 \frac{1}{x} dx$ using the trapezoidal rule with 7 ordinates.
- ii) Calculate the upper bound in the error of approximation of the integral in part (i) above using trapezoidal rule with 7 ordinates.

Solution

i)

$$f(x) = \frac{1}{x}, a = 1, b = 4, n = 6, h = \frac{b-a}{n} = \frac{4-1}{6} = \frac{1}{2}$$

	x_0	x_1	x_2	x_3	x_4	x_5	x_6
x	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	4
$f(x)$	1	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{2}{5}$	$\frac{1}{3}$	$\frac{2}{7}$	$\frac{1}{4}$
	f_0	f_1	f_2	f_3	f_4	f_5	f_6

Therefore,

$$\begin{aligned} \int_1^4 \frac{1}{x} dx &\approx \frac{h}{2} [f_0 + 2(f_1 + f_2 + f_3 + f_4 + f_5) + f_6] \\ &= \frac{1}{4} \left[1 + 2 \left(\frac{2}{3} + \frac{1}{2} + \frac{2}{5} + \frac{1}{3} + \frac{2}{7} \right) + \frac{1}{4} \right] = 1.4054 \text{ sq. units} \end{aligned}$$

- ii) The absolute error bound in using trapezoidal rule is given as

$$|E| \leq \frac{(b-a)}{12} h^2 \max_{a \leq x \leq b} |f''(x)|$$

Here, $a = 1, b = 4, n = 6, f(x) = \frac{1}{x}$. Thus, $f''(x) = \frac{2}{x^3}$ and $h = (b-a)/n = (4-1)/6 = 0.5$.
Therefore,

$$|E| \leq \frac{(4-1)}{12} (0.5)^2 \max_{1 \leq x \leq 4} \left| \frac{2}{x^3} \right| = \frac{3}{12} (0.5)^2 \left| \frac{2}{(1)^3} \right| = 0.125$$

Exercise:

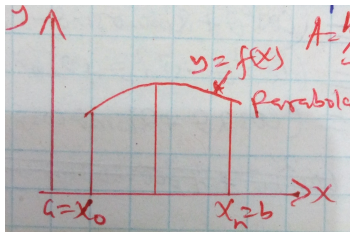
1. Use trapezoidal rule to estimate $\int_1^2 \frac{5x}{2 + \ln 5x} dx$ correct to 4dp using 5 ordinates.
2. Evaluate $\int_2^6 \frac{x}{1+x^2} dx$ correct to 4dp using Trapezoidal rule with 8 subdivisions.

7.3 Simpson's rule

This approximation procedure uses parabolic arcs rather than line segments to approximate portions of the graph $y = f(x)$. It uses the fact that if x_0, x_1 and x_2 are three consecutive ordinates, then the definite integral of the quadratic function passing through the points $(x_0, y_0), (x_1, y_1)$ and (x_2, y_2) is equal to $\frac{h}{3}(y_0 + 4y_1 + y_2)$. We divide the interval $[a, b]$ into n subintervals, but now we must require that n must be even. We then approximate the given integral over each pair of subintervals by expression (i) to get

$$\begin{aligned} \int_a^b f(x) dx &\approx \frac{h}{3} [(f_0 + 4f_1 + f_2) + (f_2 + 4f_3 + f_4) + \cdots + (f_{n-2} + 4f_{n-1} + f_n)] \\ &= \frac{h}{3} [f_0 + 4(f_1 + f_3 + \cdots + f_{n-1}) + 2(f_2 + f_4 + \cdots + f_{n-2}) + f_n] \end{aligned}$$

This is called the Simpson's rule.

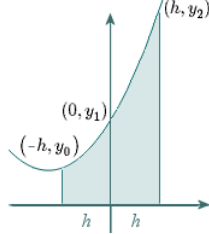


→ Note: The bound for the truncation error (E) in using the Simpson's rule is given as

$$|E| \leq \frac{(b-a)}{180} h^4 \max_{a \leq x \leq b} |f^{(4)}(x)|$$

□ Prove Simpson's rule (**Assignment**)

We consider the area under the general parabola $f(x) = ax^2 + bx + c$ between $x = -h$ and $x = h$ as shown below.



We have:

$$A_1 = \int_{-h}^h f(x) dx \approx \int_{-h}^h (ax^2 + bx + c) dx = \left[a \frac{x^3}{3} + b \frac{x^2}{2} + cx \right]_{-h}^h = \frac{h}{3} [2ah^2 + 6c]$$

Our parabola passes through the points $(-h, y_0)$, $(0, y_1)$ and (h, y_2) . Substituting these x and y values into the general equation of our parabola, we get: $y_0 = ah^2 - bh + c$, $y_1 = c$, $y_2 = ah^2 + bh + c \Rightarrow y_0 - 2y_1 + y_2 = 2ah^2$. Therefore,

$$A_1 = \int_{-h}^h f(x) dx = \frac{h}{3} [y_0 - 2y_1 + y_2 + 6y_1] = \frac{h}{3} [y_0 + 4y_1 + y_2]$$

The parabola passing through the next set of three points will have an area of

$$A_2 = \frac{h}{3} [y_2 + 4y_3 + y_4]$$

Thus,

$$A_1 + A_2 = \frac{h}{3} [y_0 + 4(y_1 + y_3) + 2y_2 + y_4]$$

We could keep going by creating more and more segments, and adding the areas as we go along and we would obtain the Simpson's rule above.

Example(s):

1. Approximate $\int_1^4 \frac{1}{x} dx$ using the Simpson's rule with 7 ordinates.

Solution

$$f(x) = \frac{1}{x}, a = 1, b = 4, n = 6, h = \frac{b-a}{n} = \frac{4-1}{6} = \frac{1}{2}$$

	x_0	x_1	x_2	x_3	x_4	x_5	x_6
x	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	4
$f(x)$	1	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{2}{5}$	$\frac{1}{3}$	$\frac{2}{7}$	$\frac{1}{4}$
	f_0	f_1	f_2	f_3	f_4	f_5	f_6

Therefore,

$$\begin{aligned} \int_1^4 \frac{1}{x} dx &\approx \frac{h}{2} [f_0 + 4(f_1 + f_3 + f_5) + 2(f_2 + f_4) + f_6] \\ &= \frac{1}{6} \left[1 + 4 \left(\frac{2}{3} + \frac{2}{5} + \frac{2}{7} \right) + 2 \left(\frac{1}{2} + \frac{1}{3} \right) + \frac{1}{4} \right] = 1.3877 \text{ sq. units} \end{aligned}$$

Exercise:

1. Use Simpson's rule to approximate the composite integral $\int_0^1 \frac{1}{1+x} dx$ correct to 4dp (Take $h = 0.125$). [9 Marks]
2. Use Simpson's rule to approximate the value of the integral $\int_1^5 \frac{7 \cos 2x}{x} dx$ using $n = 4$. Give your function values correct to 5dp.